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(54) **MOUNTING RAILS**

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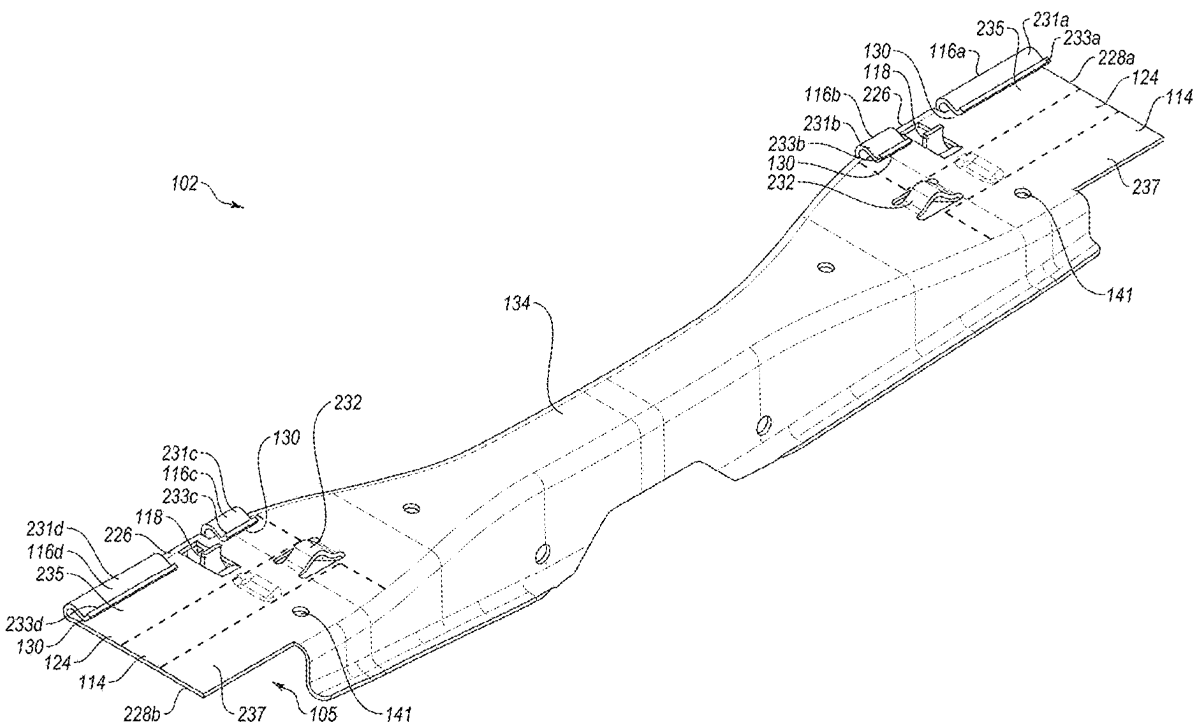
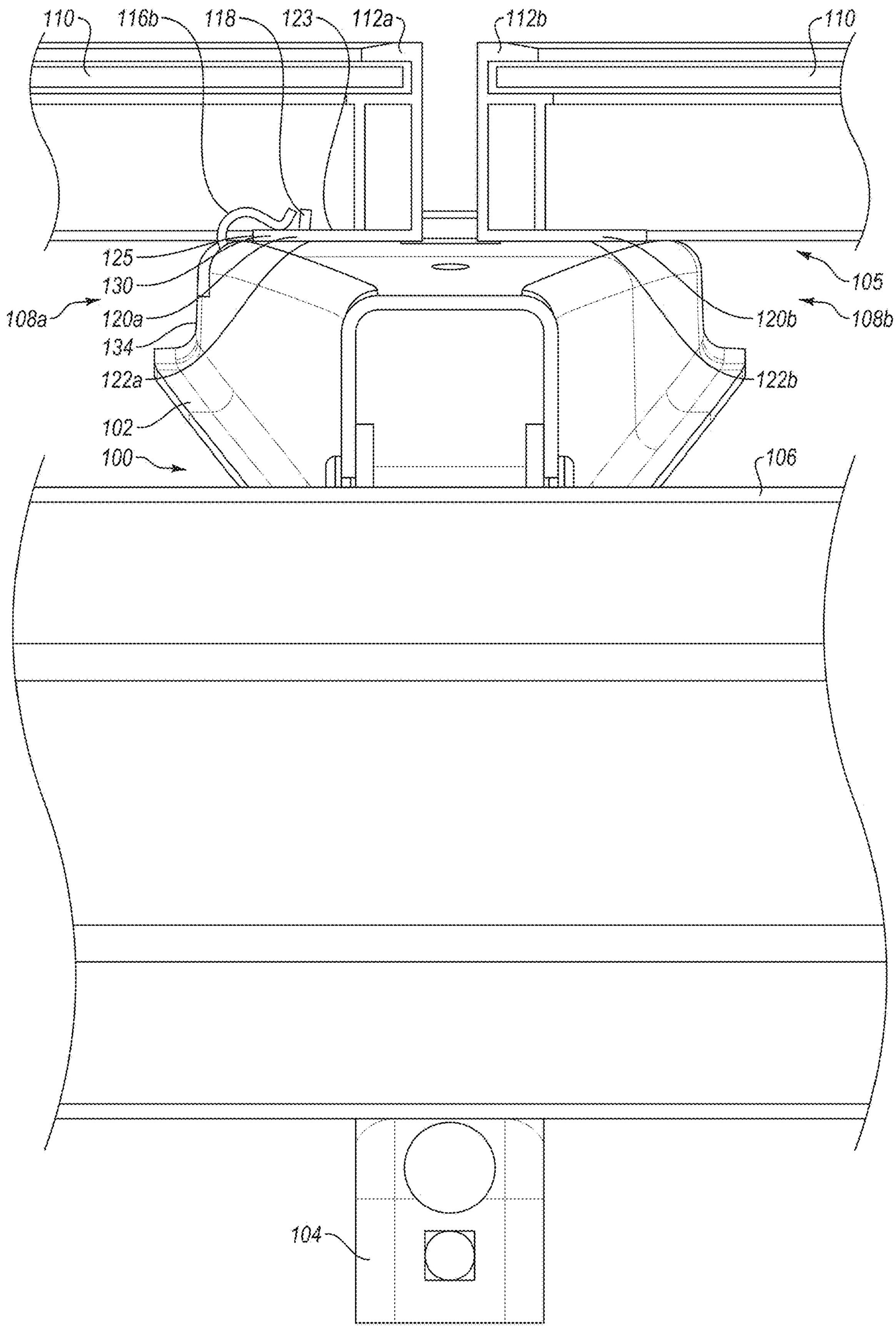
(57) **ABSTRACT**

A mounting rail may be configured to couple to a first photovoltaic (PV) module and a second PV module. The mounting rail may include a hooked mechanism and an attachment feature. The hooked mechanism may at least partially define an aperture configured to receive a portion of a first module frame associated with the first PV module. The hooked mechanism may also physically engage with a surface of the first module frame to couple the first module frame to the mounting rail and to prevent the first module frame from unintentionally uncoupling from the mounting rail. The attachment feature may interface with a second module frame associated with the second PV module to couple the second module frame to the mounting rail.

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106
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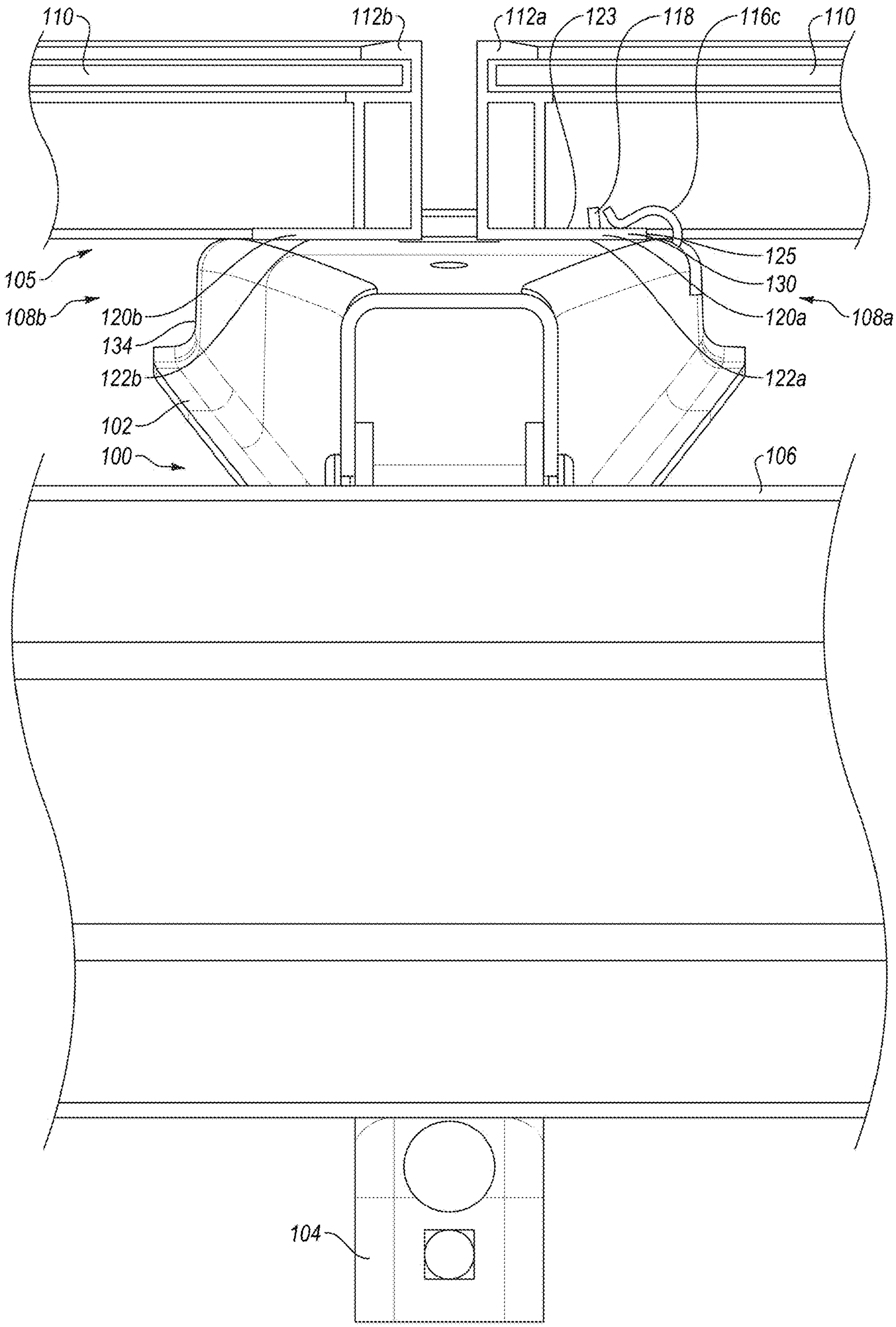
FIG. 1A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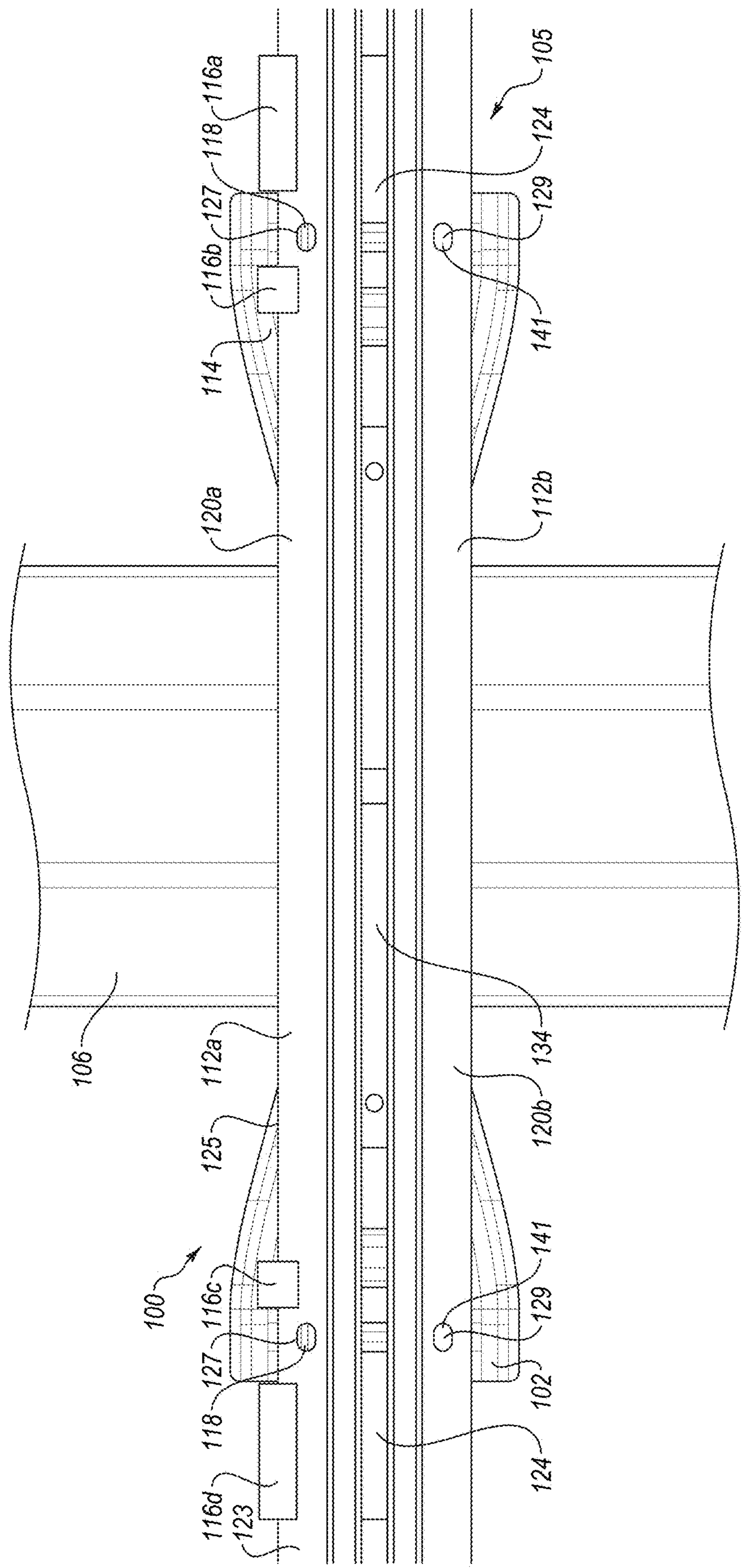
106
104

FIG. 1B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116c
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116d
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112b
134
120b
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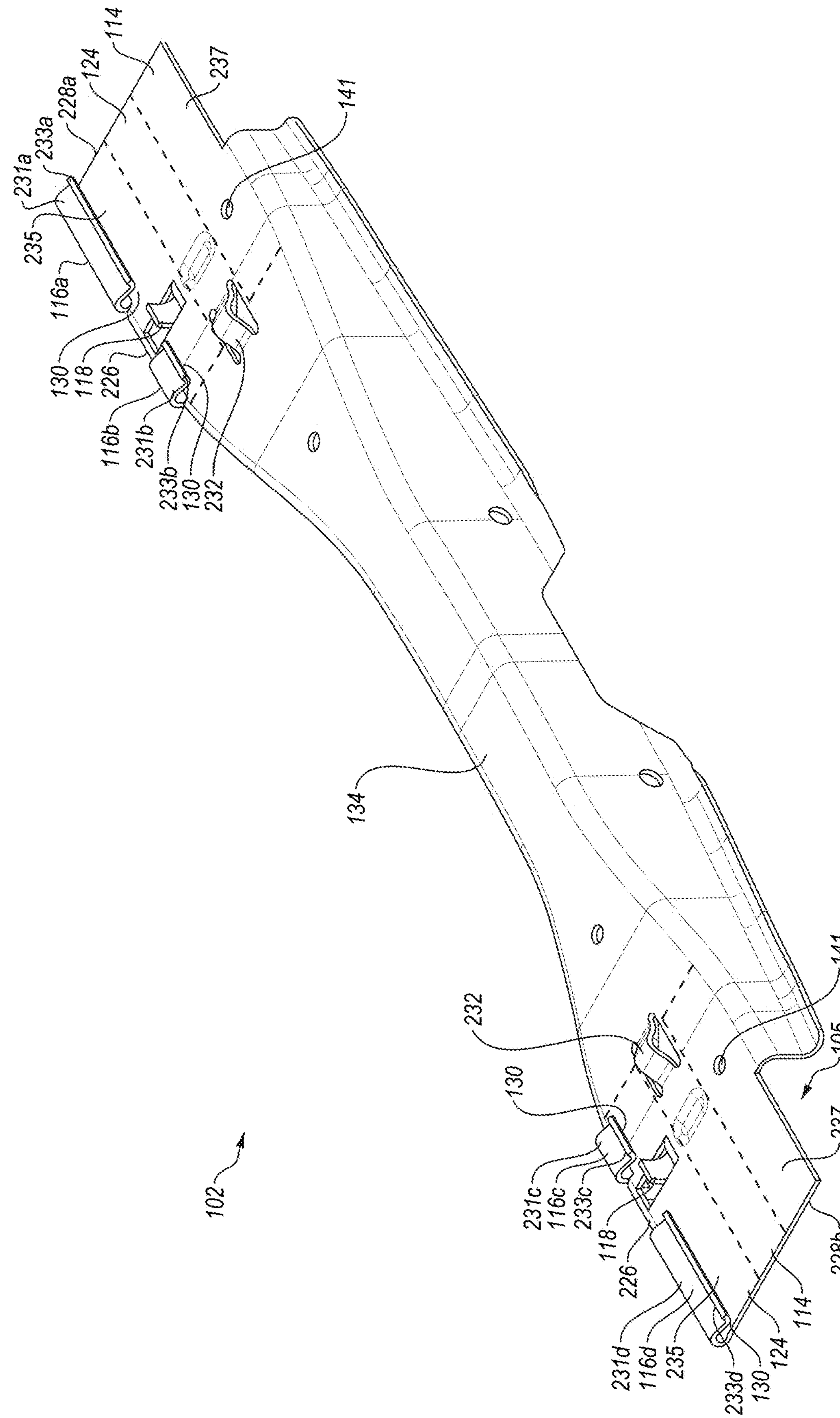
129
102
124

FIG. 1C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116a
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232
130
231c
116c
233c
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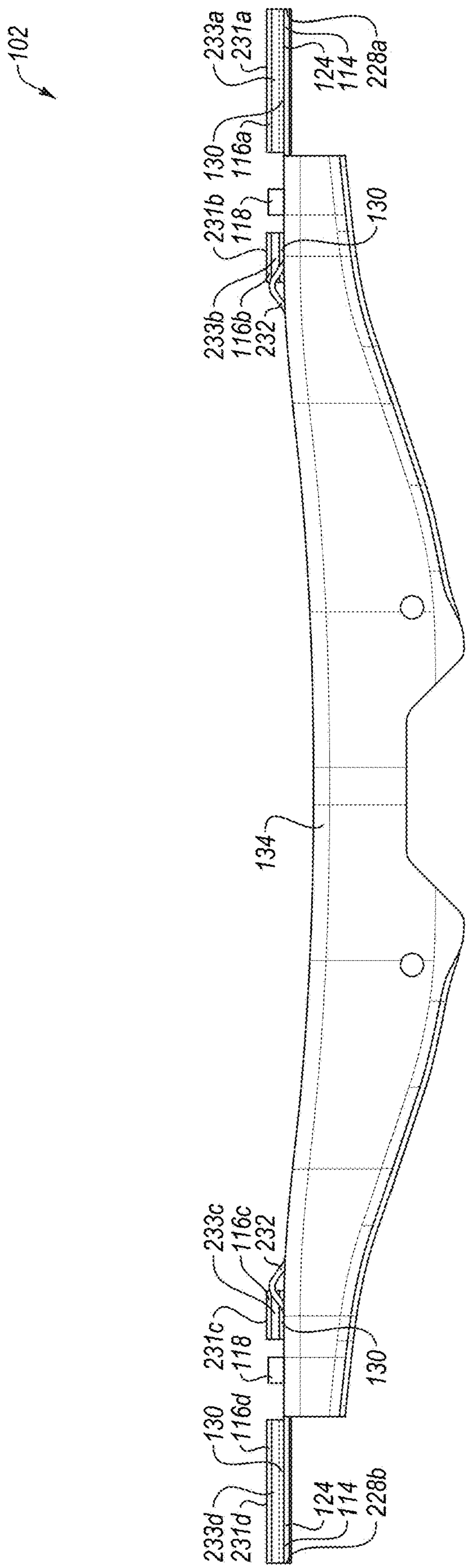
226
231d
116d
235
233d
130
124
114
228b
237
105
141
102

FIG. 2A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233b
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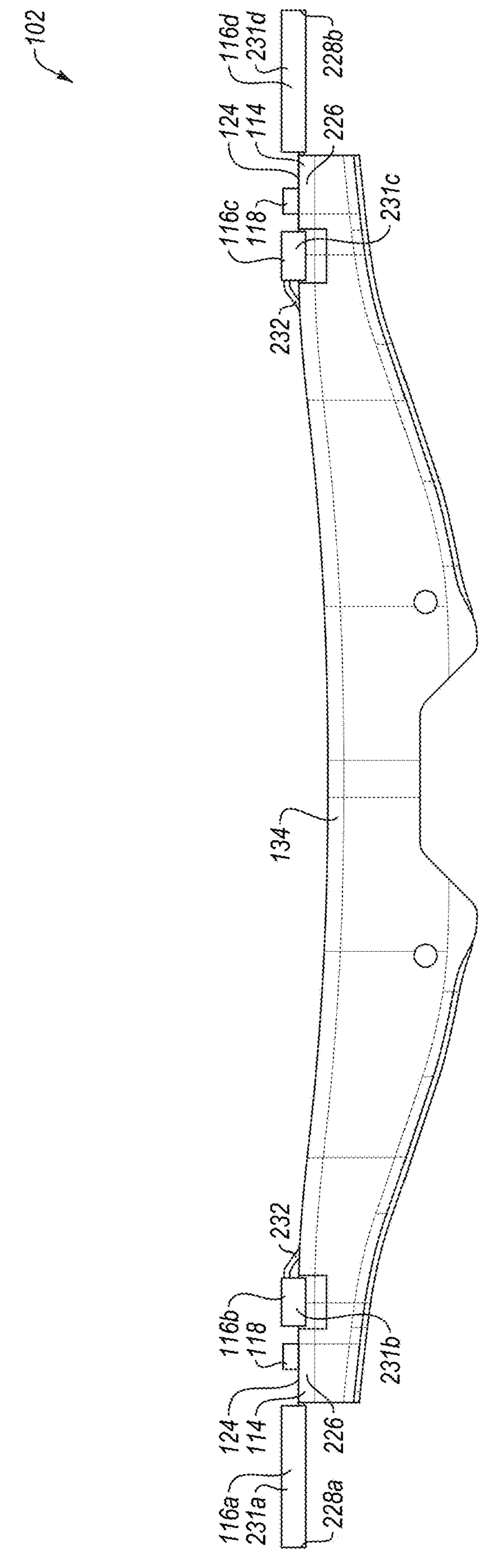
116d
233d
231d
124
114
228b

FIG. 2B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114
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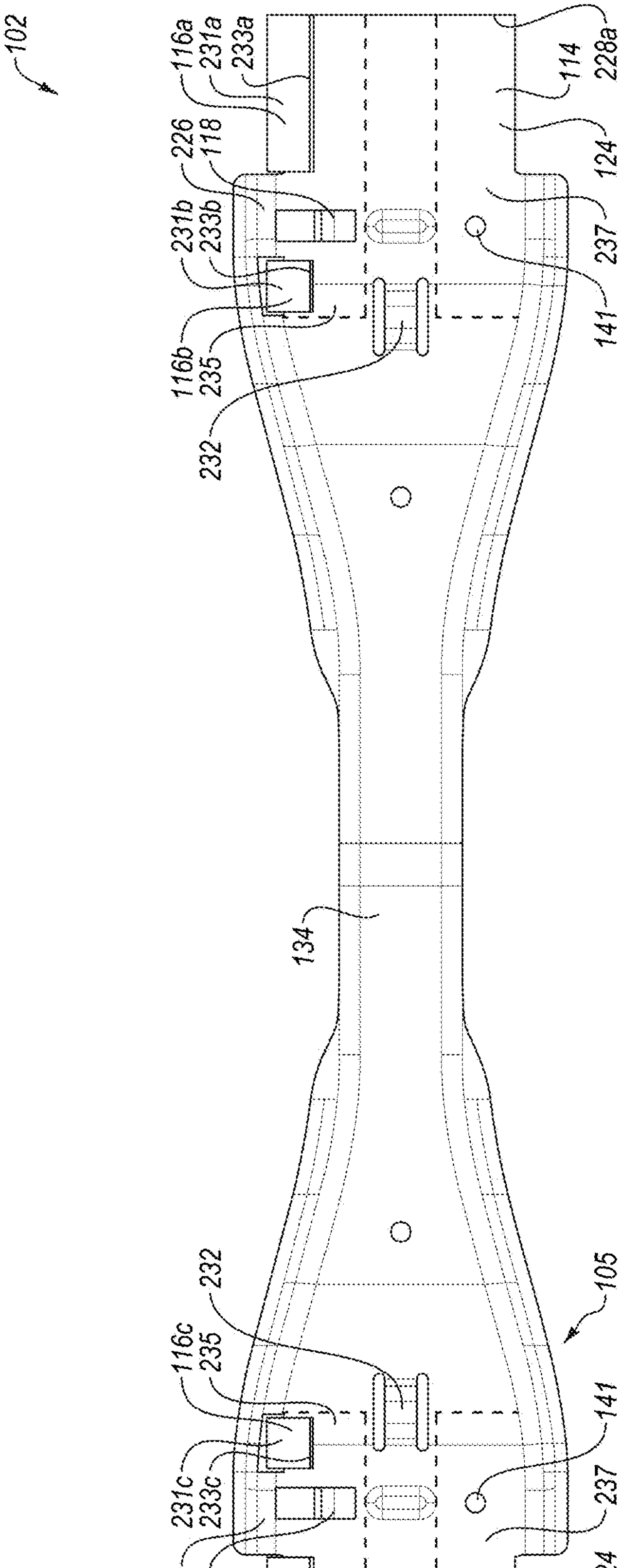
231a
228a

FIG. 2C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228b

FIG. 2D

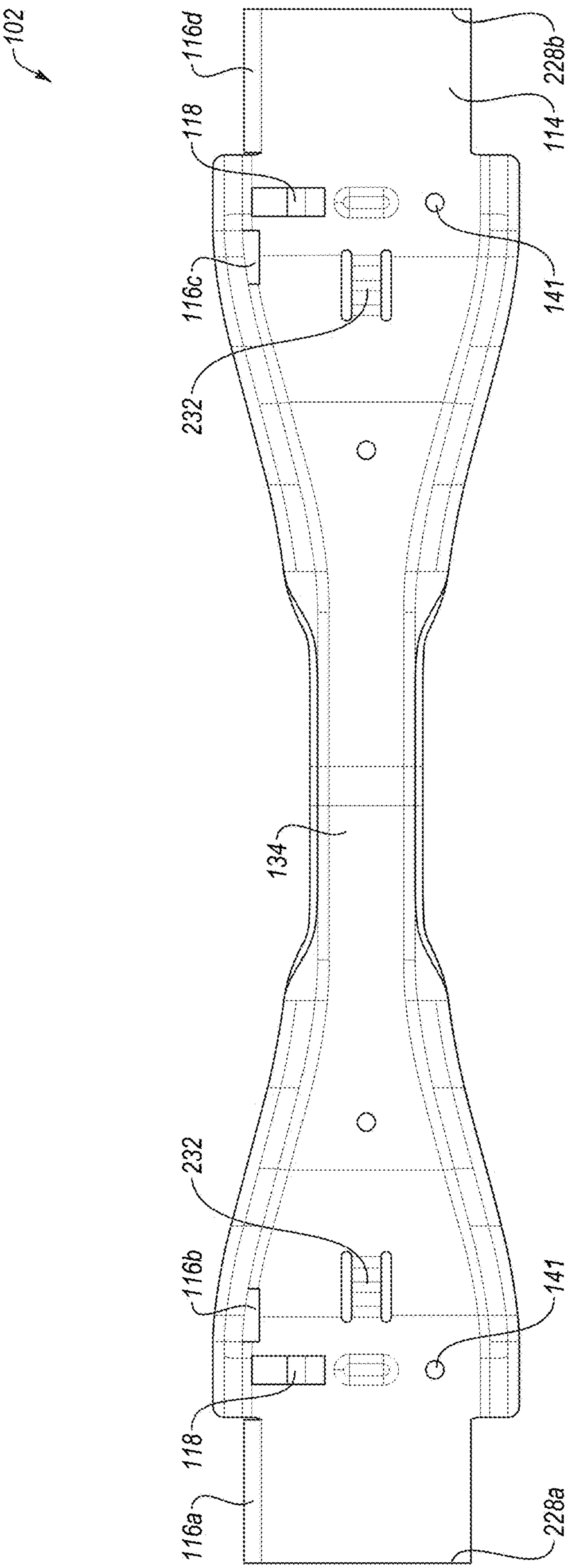
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FIG. 2E

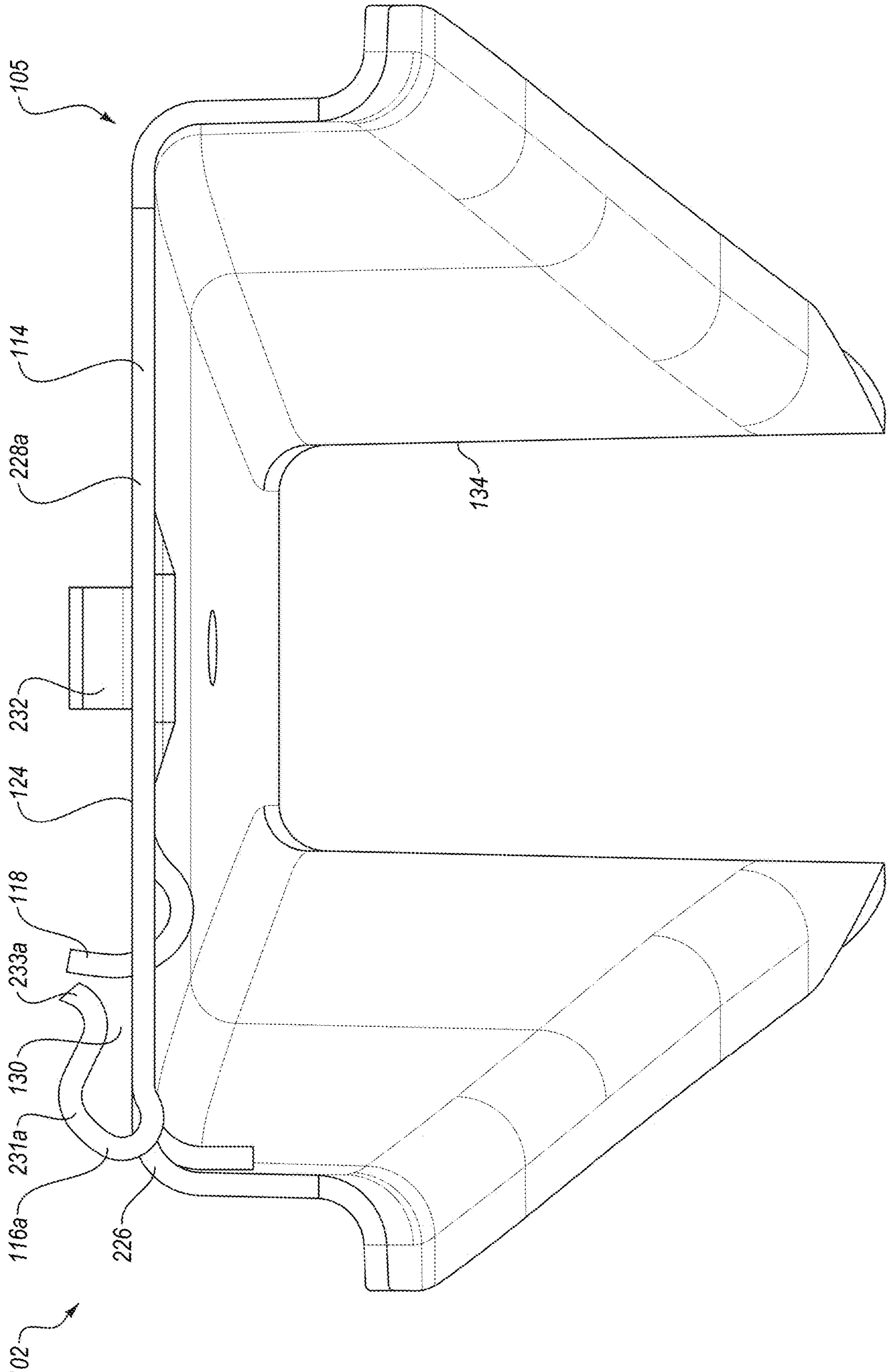


FIG. 2F

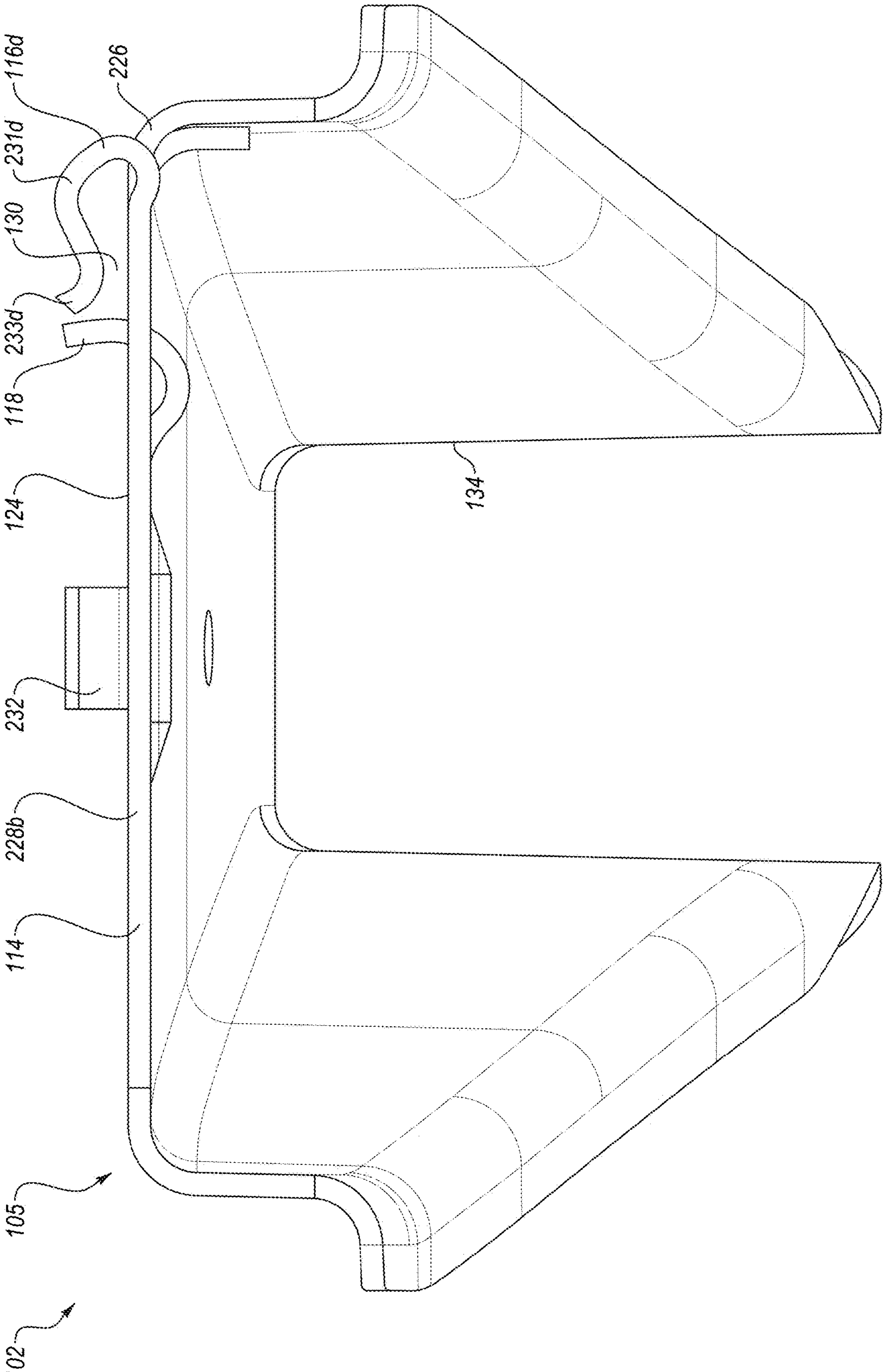


FIG. 2G

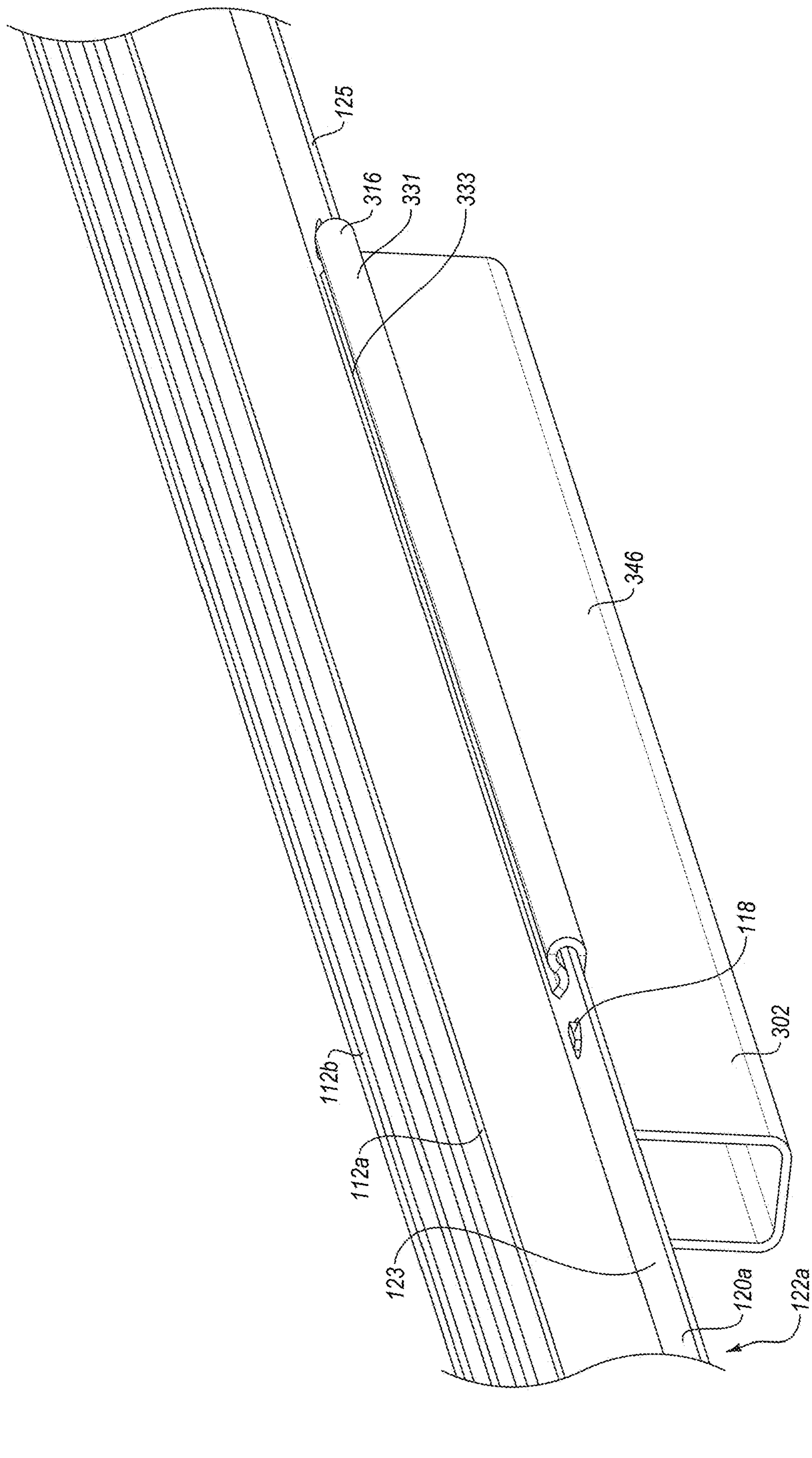
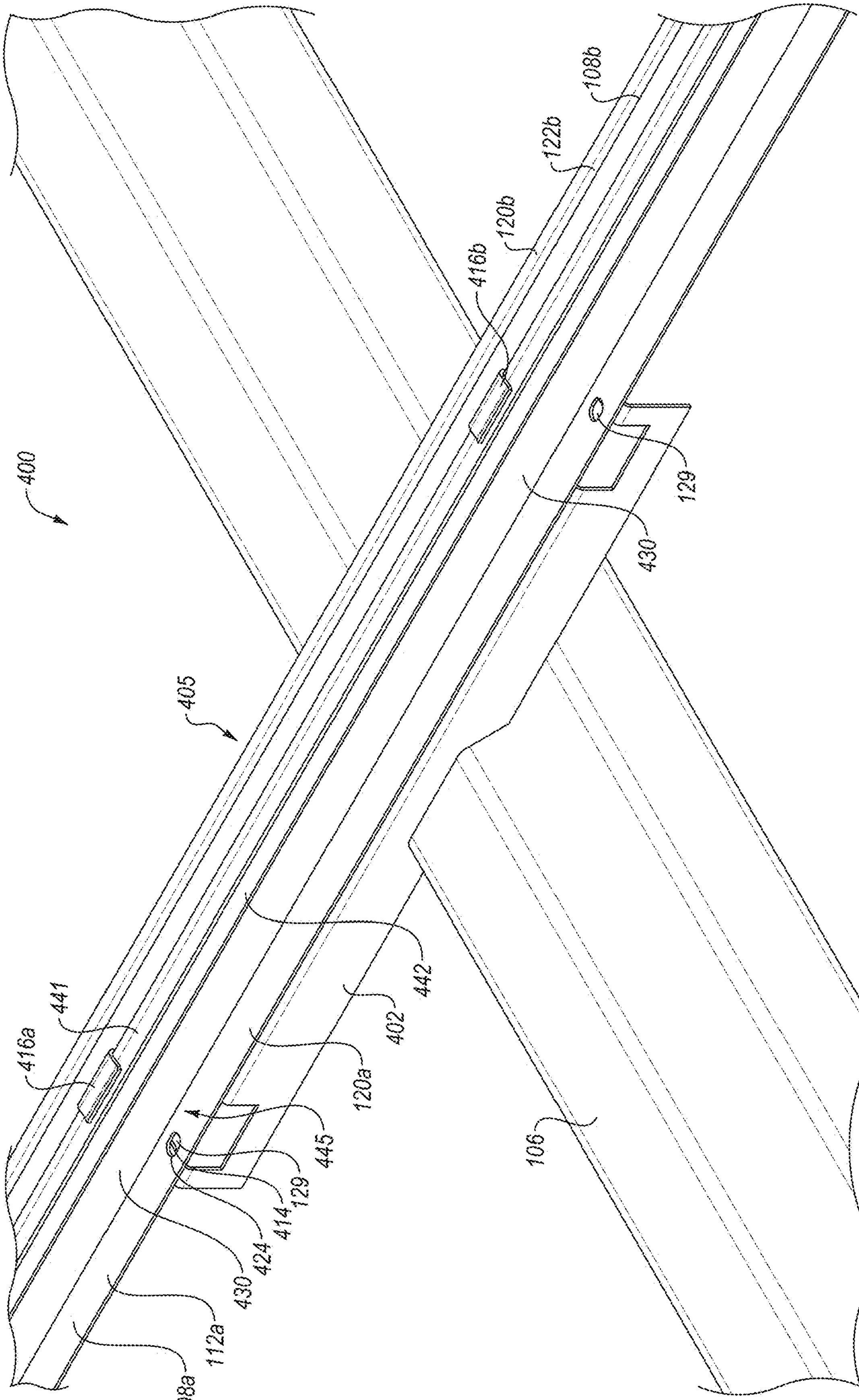


FIG. 3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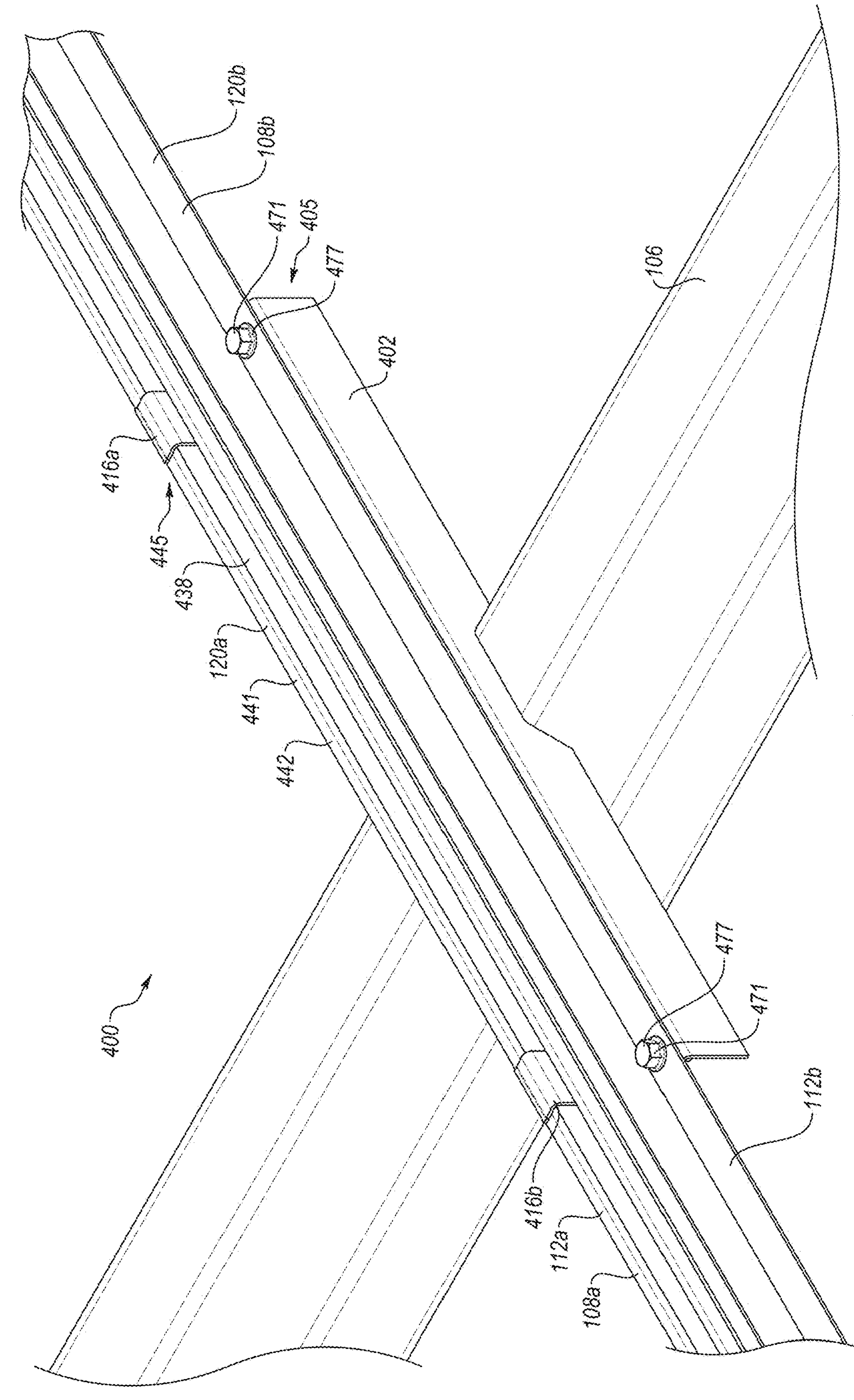
112a
108a
106

FIG. 4A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112a
108a

FIG. 4B

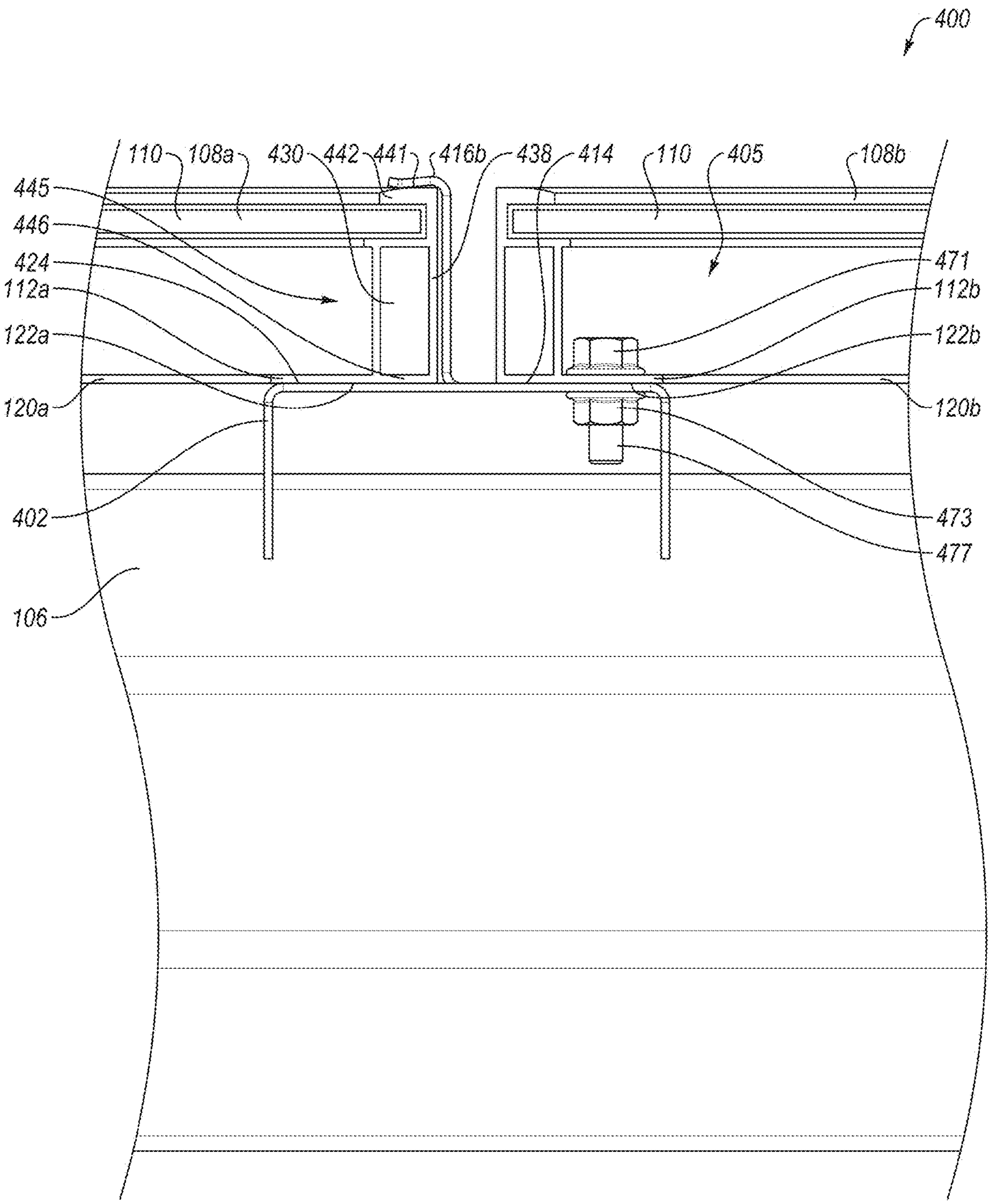
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FIG. 4C

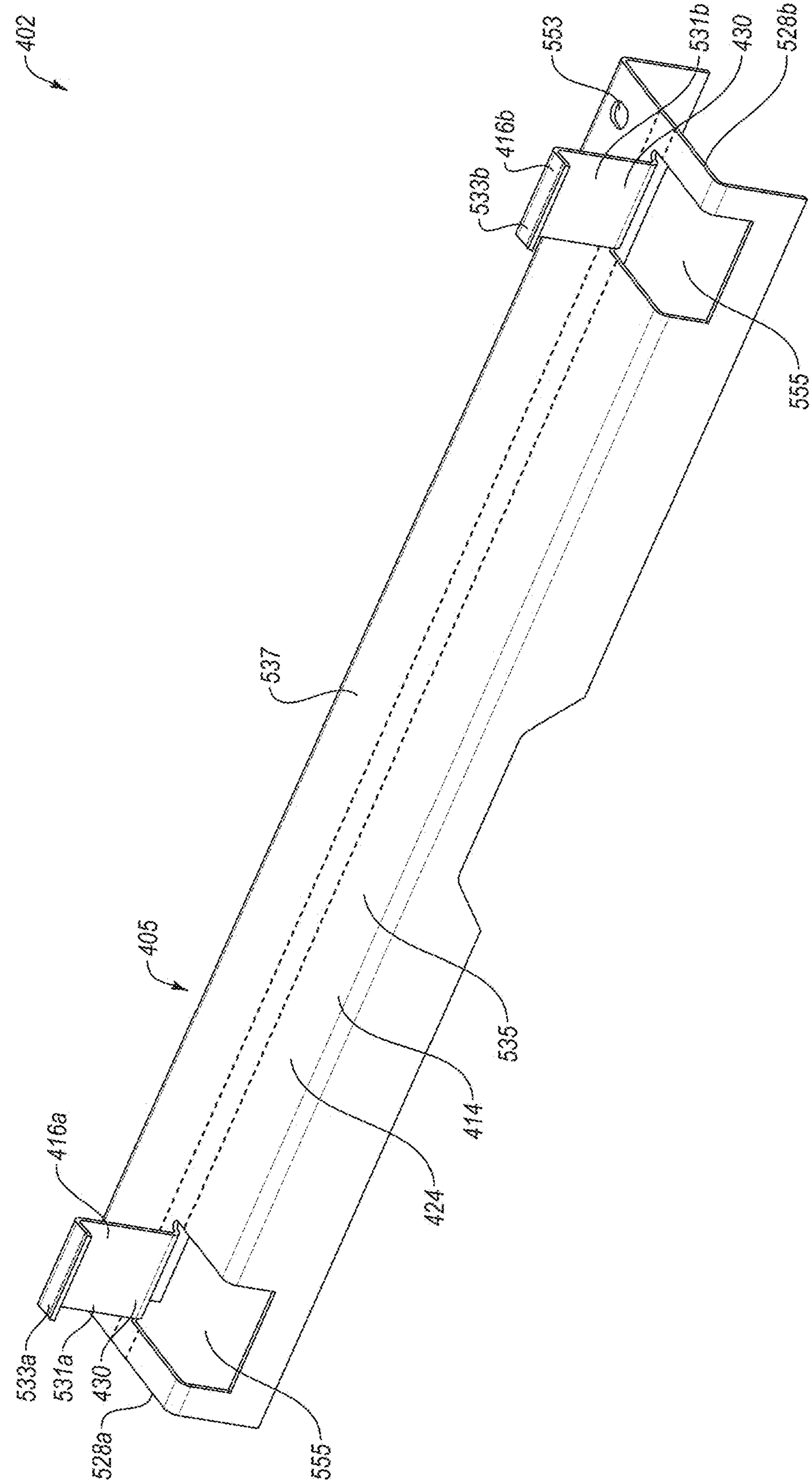
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FIG. 5A

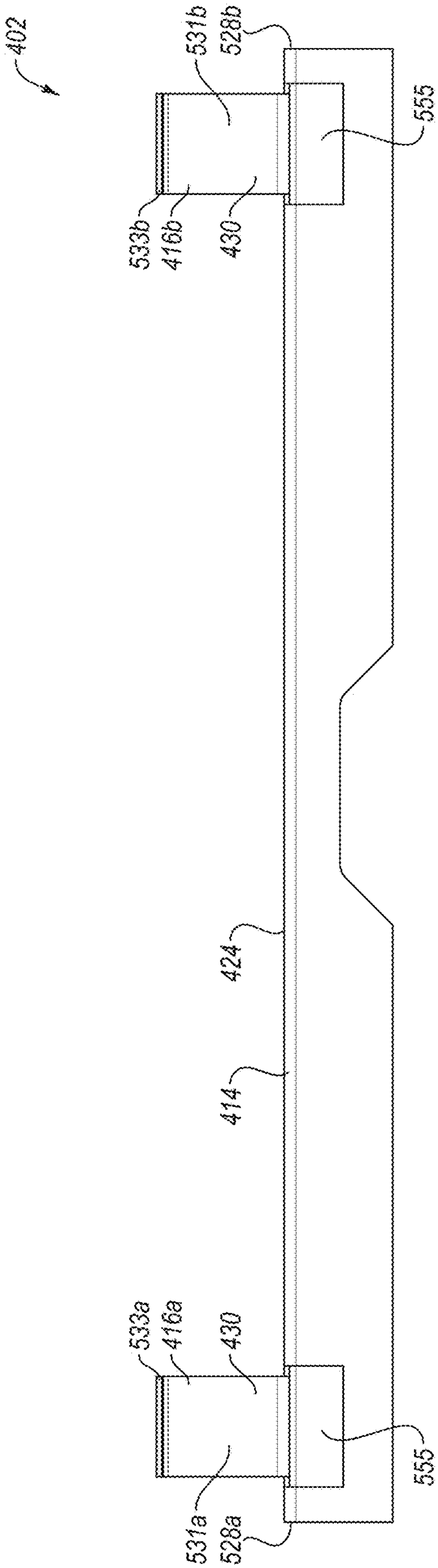
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FIG. 5B

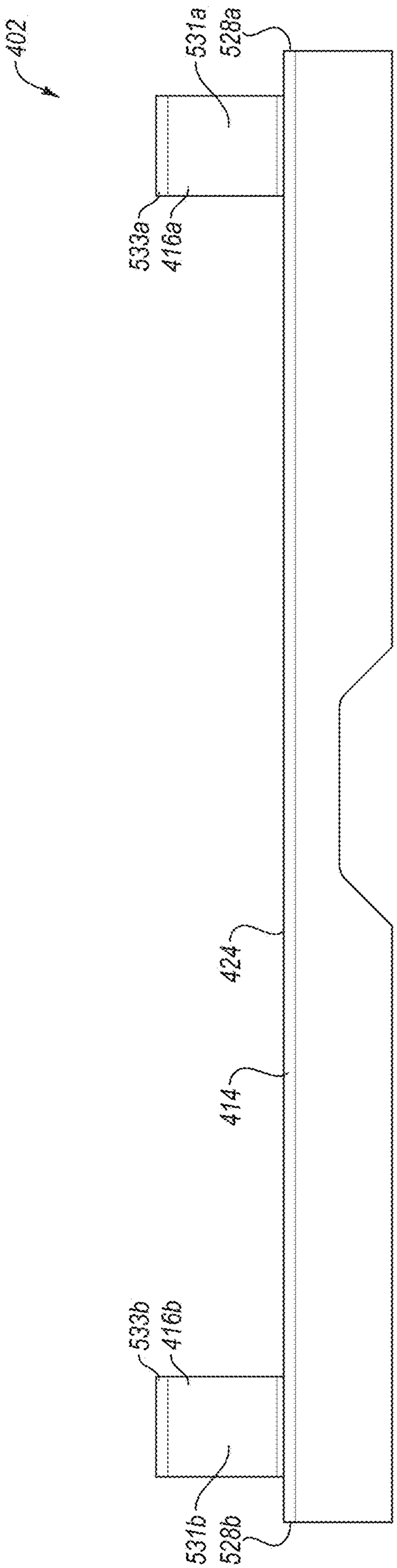

FIG. 5C

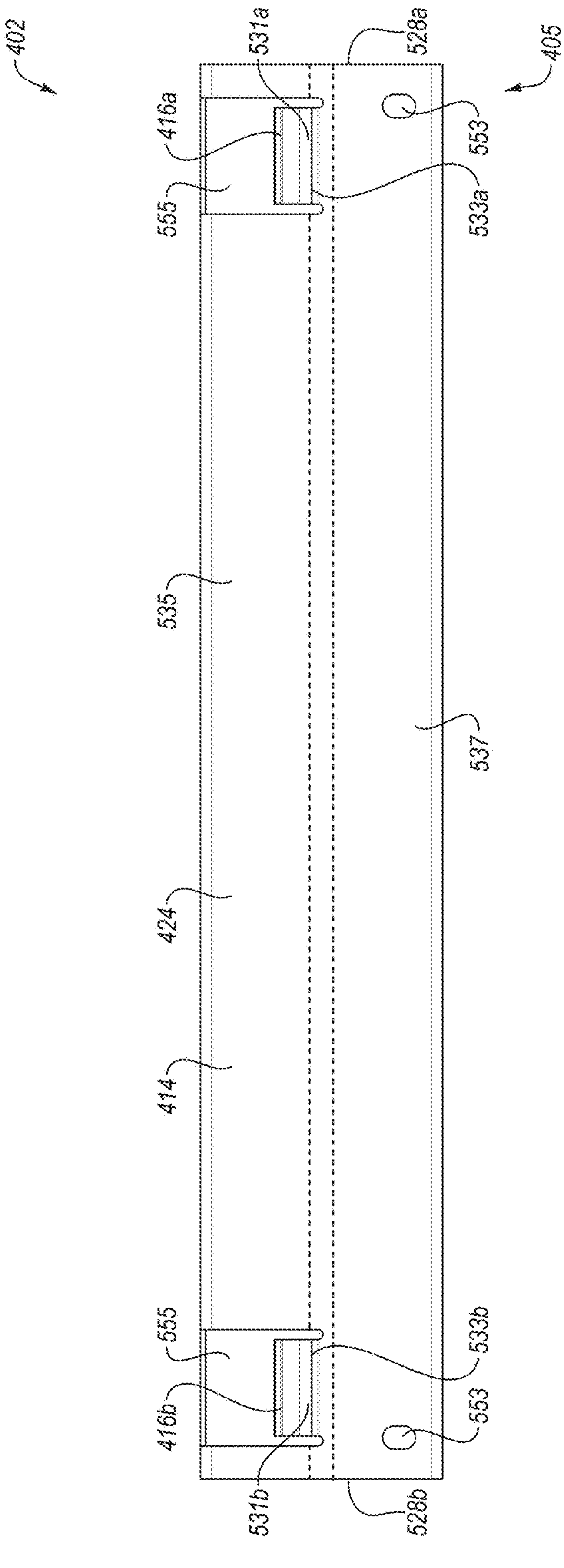
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FIG. 5D

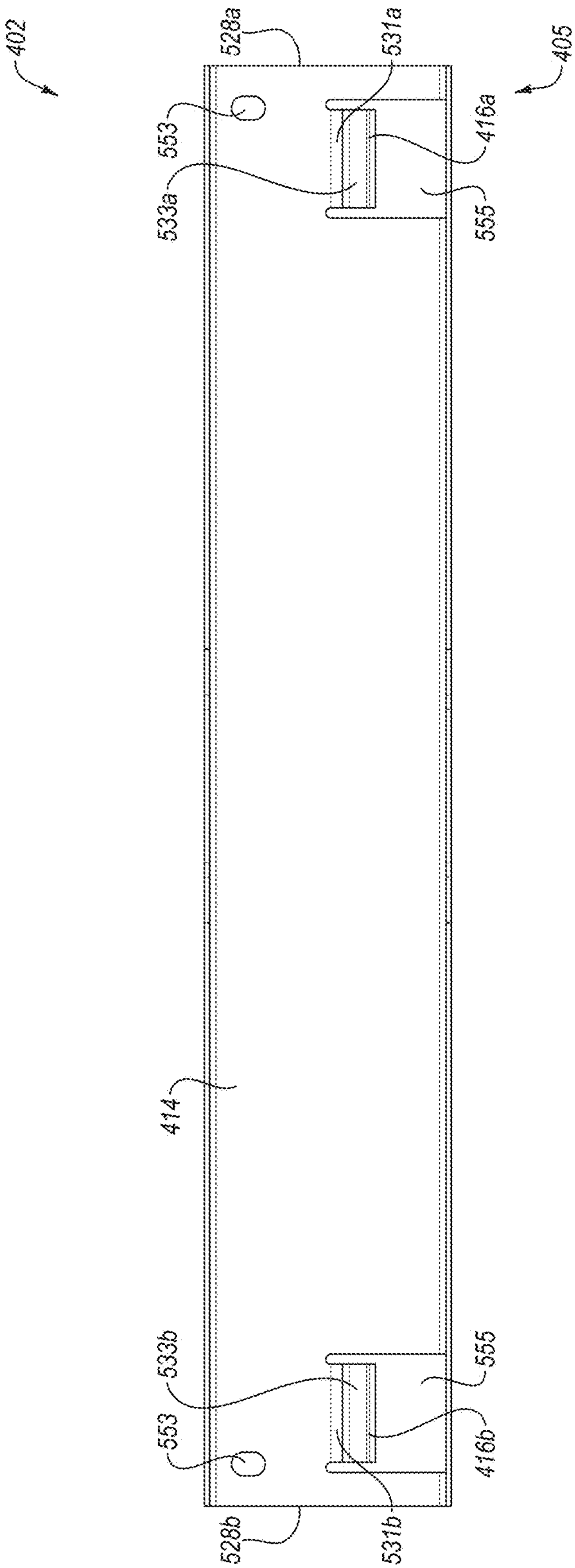
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FIG. 5E

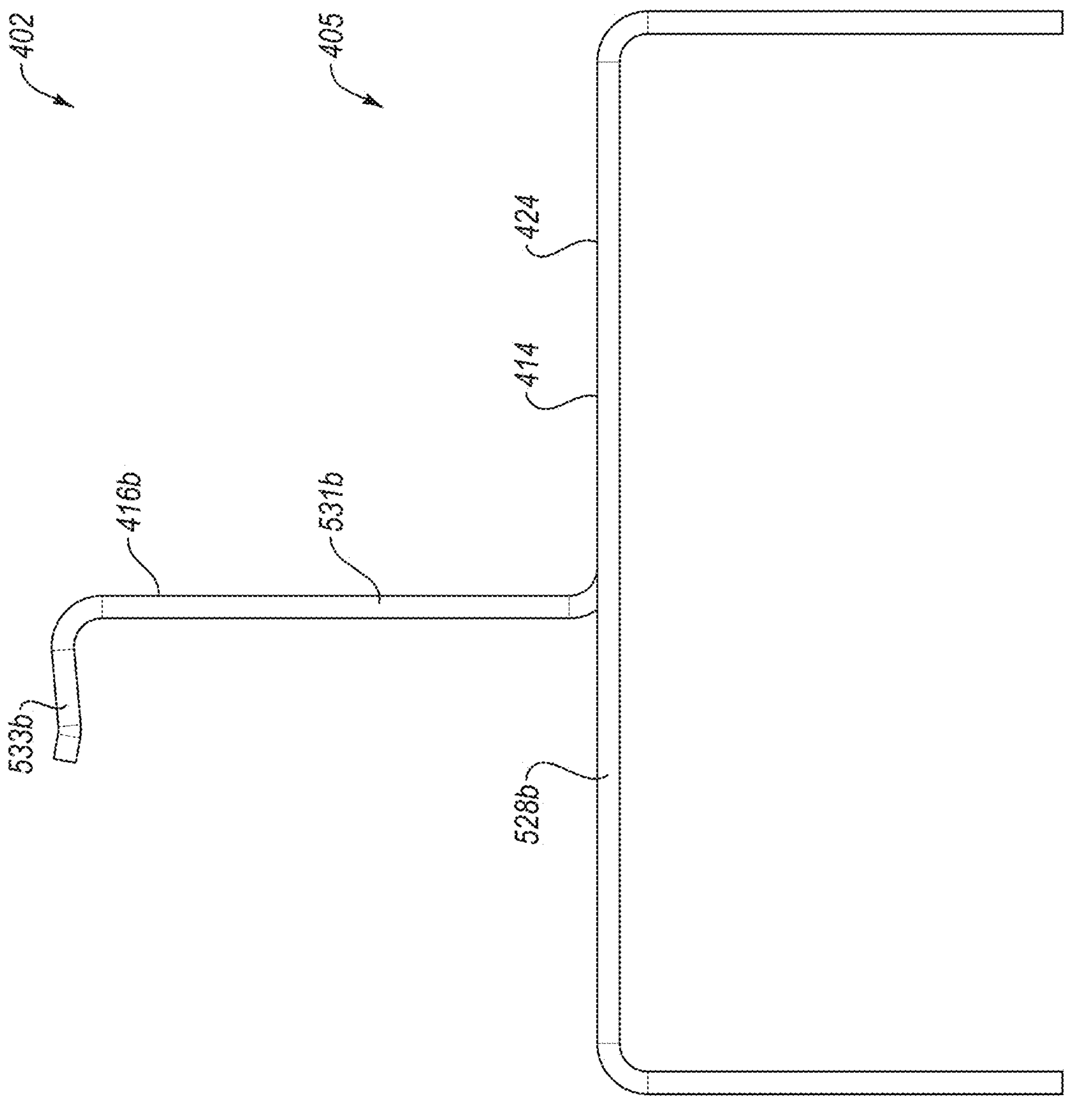


FIG. 5F

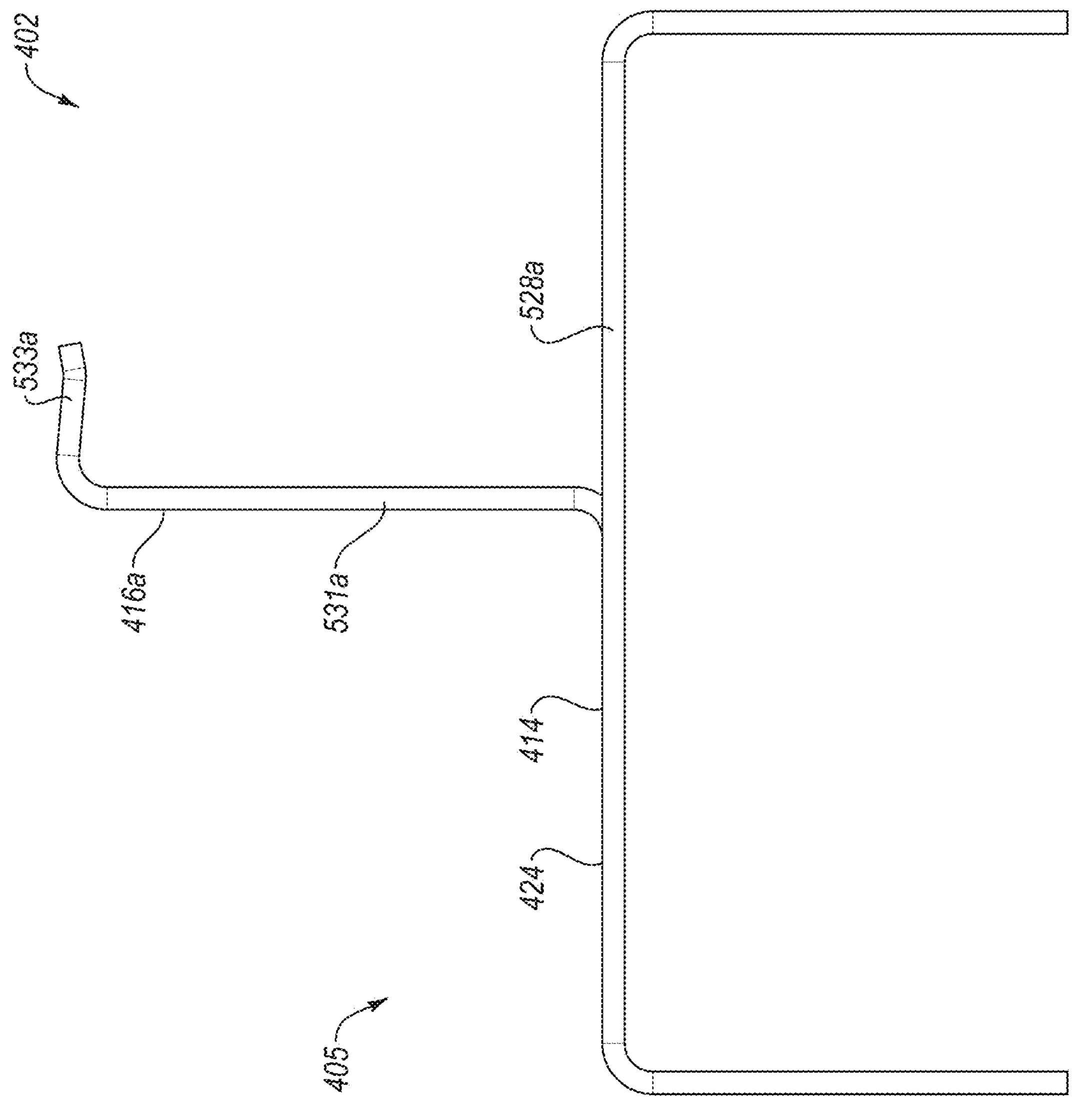
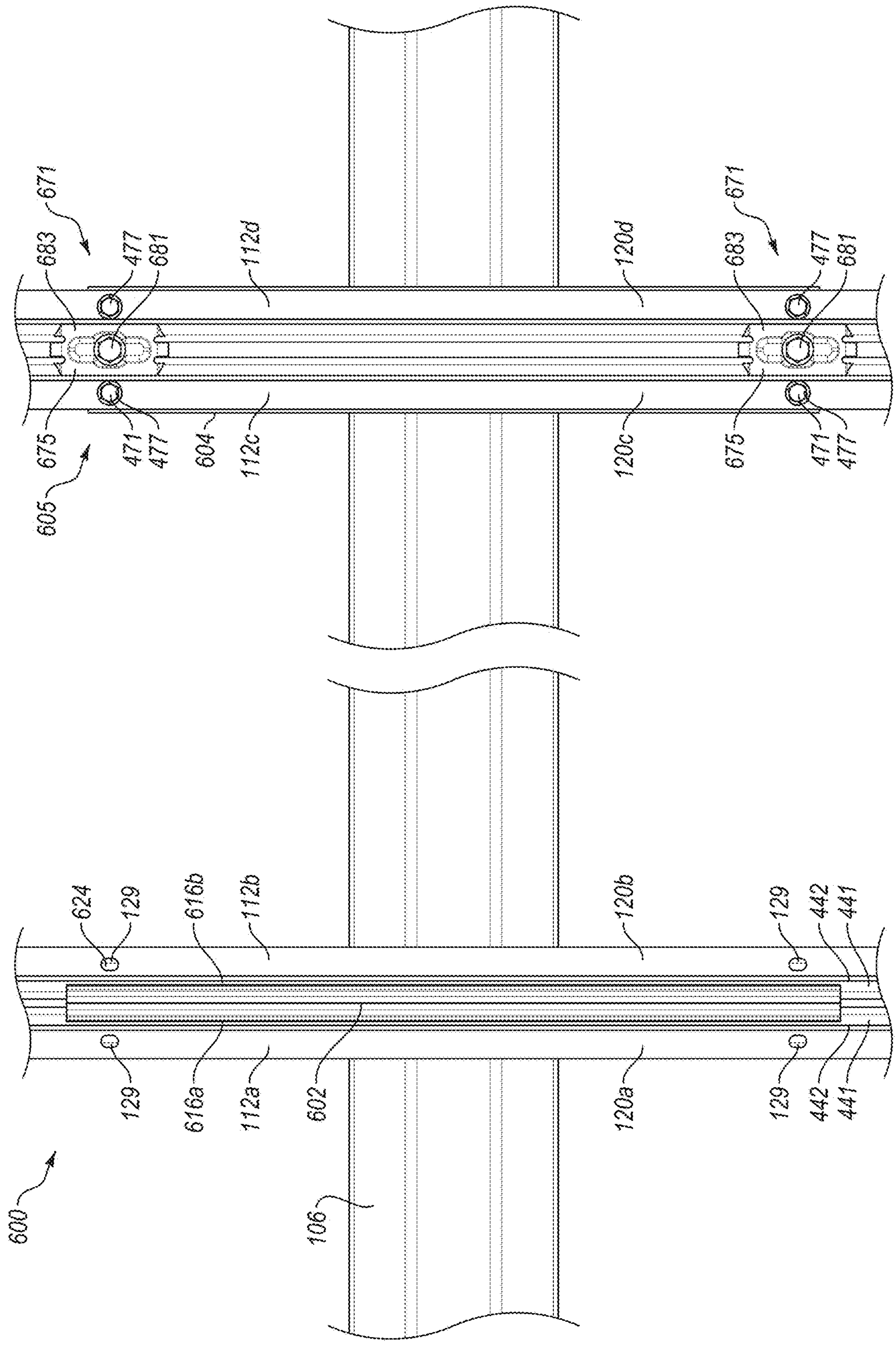


FIG. 5G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675
471
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120d
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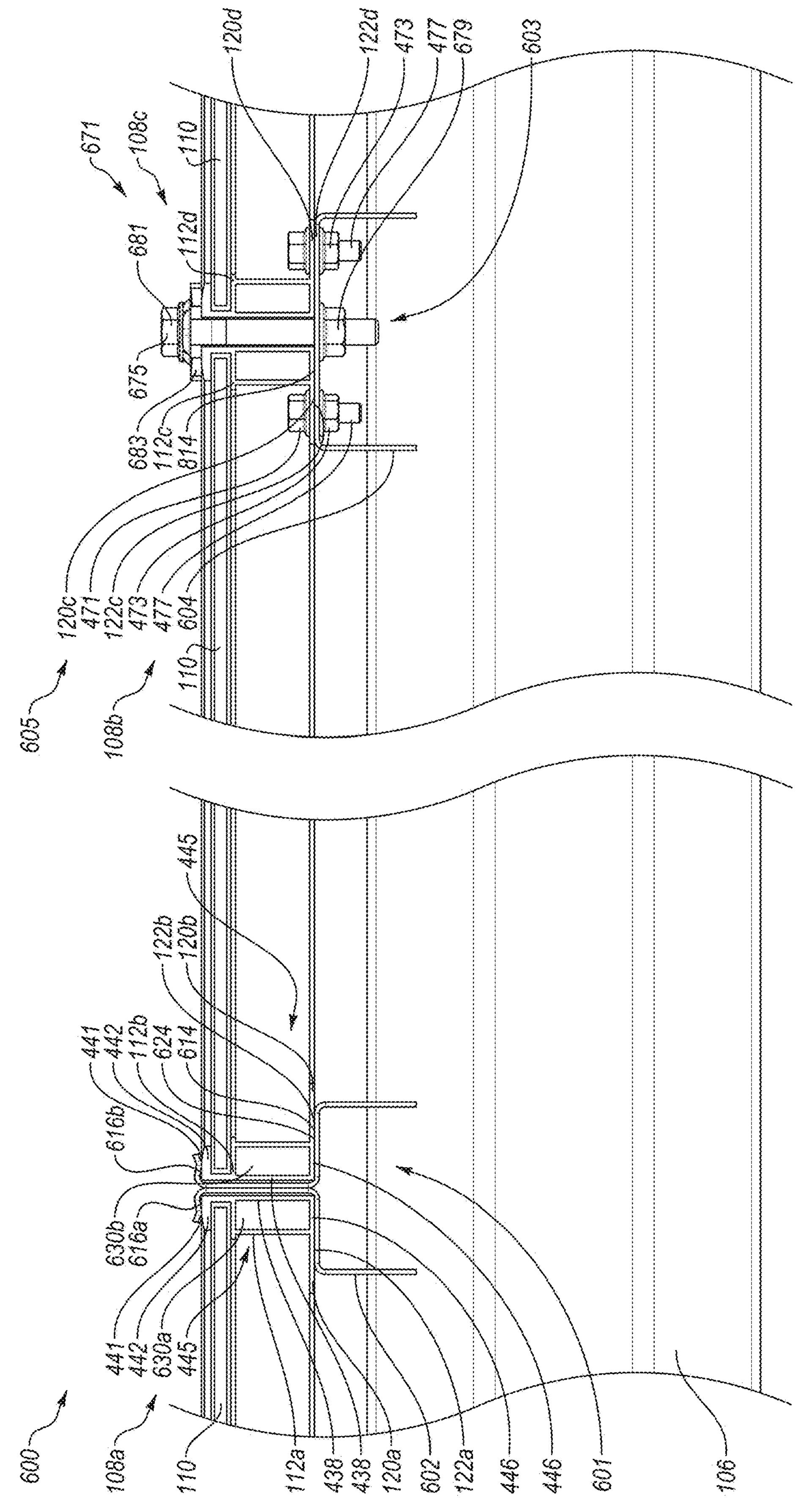
671
477
681

FIG. 6A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112b
624
614
122b
120b
445
605
108b
110
120c
471
122c
473
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603

FIG. 6B

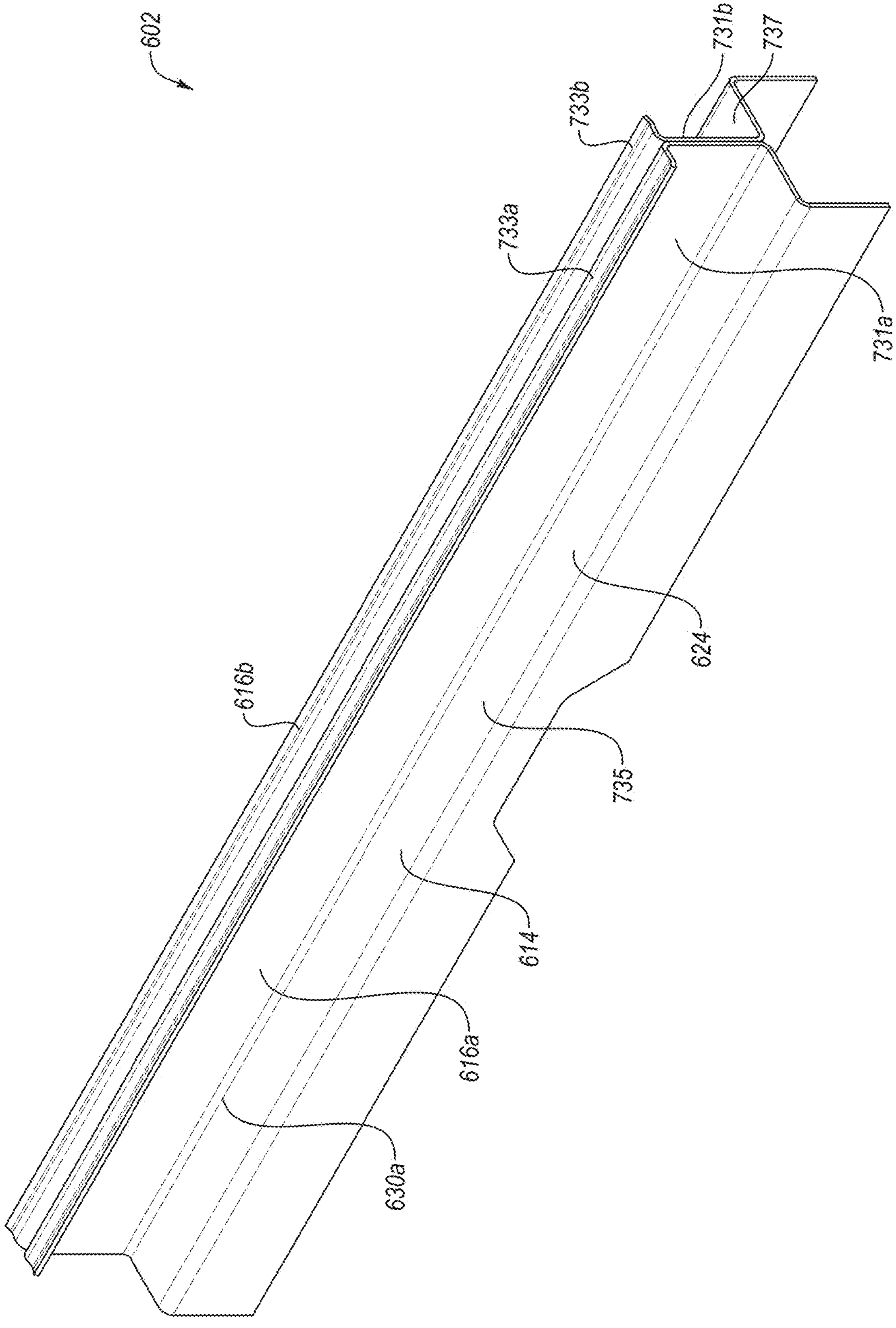


FIG. 7A

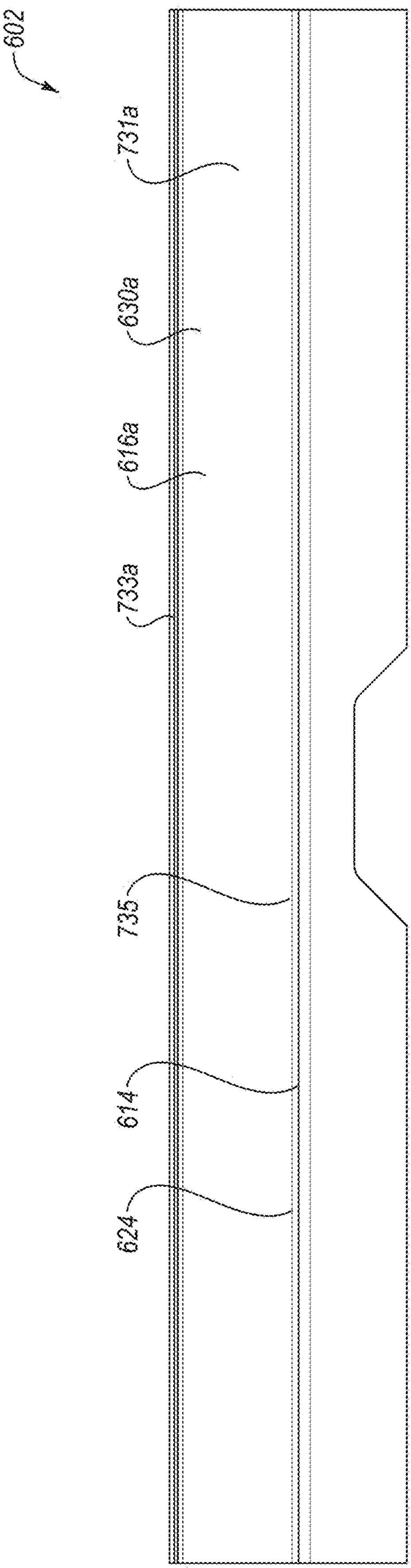


FIG. 7B

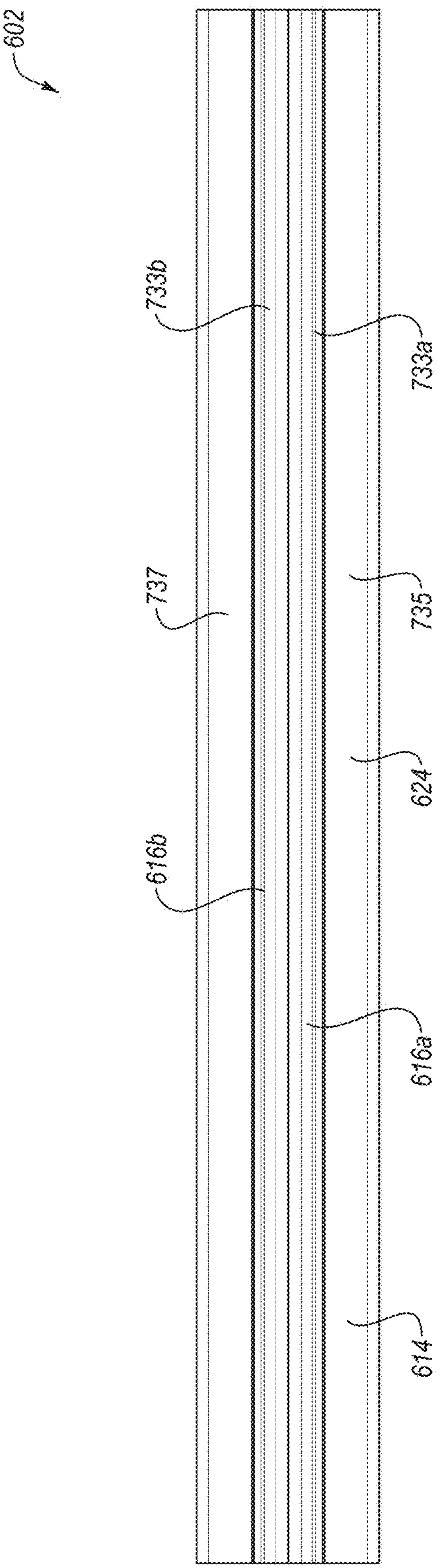


FIG. 7C

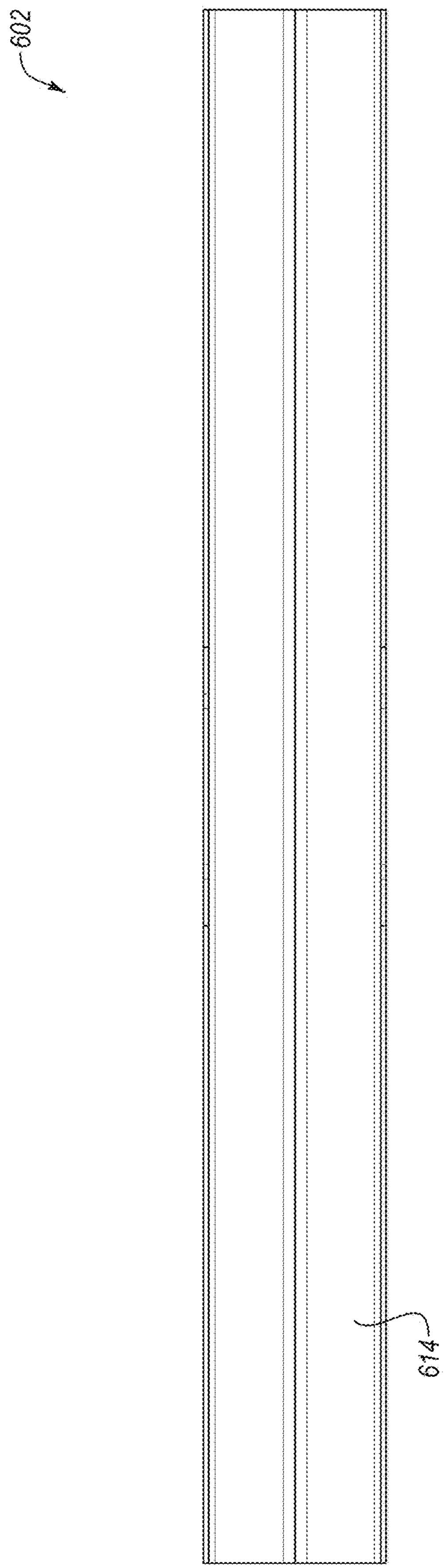


FIG. 7D

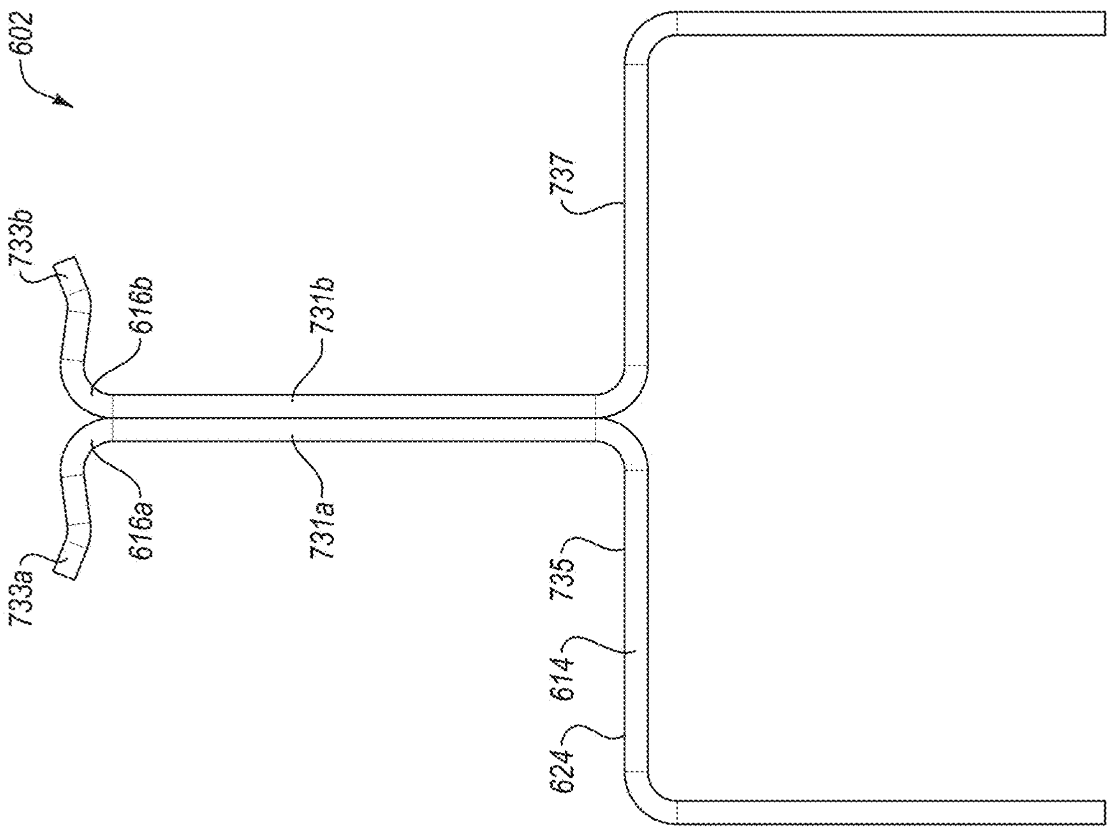


FIG. 7E

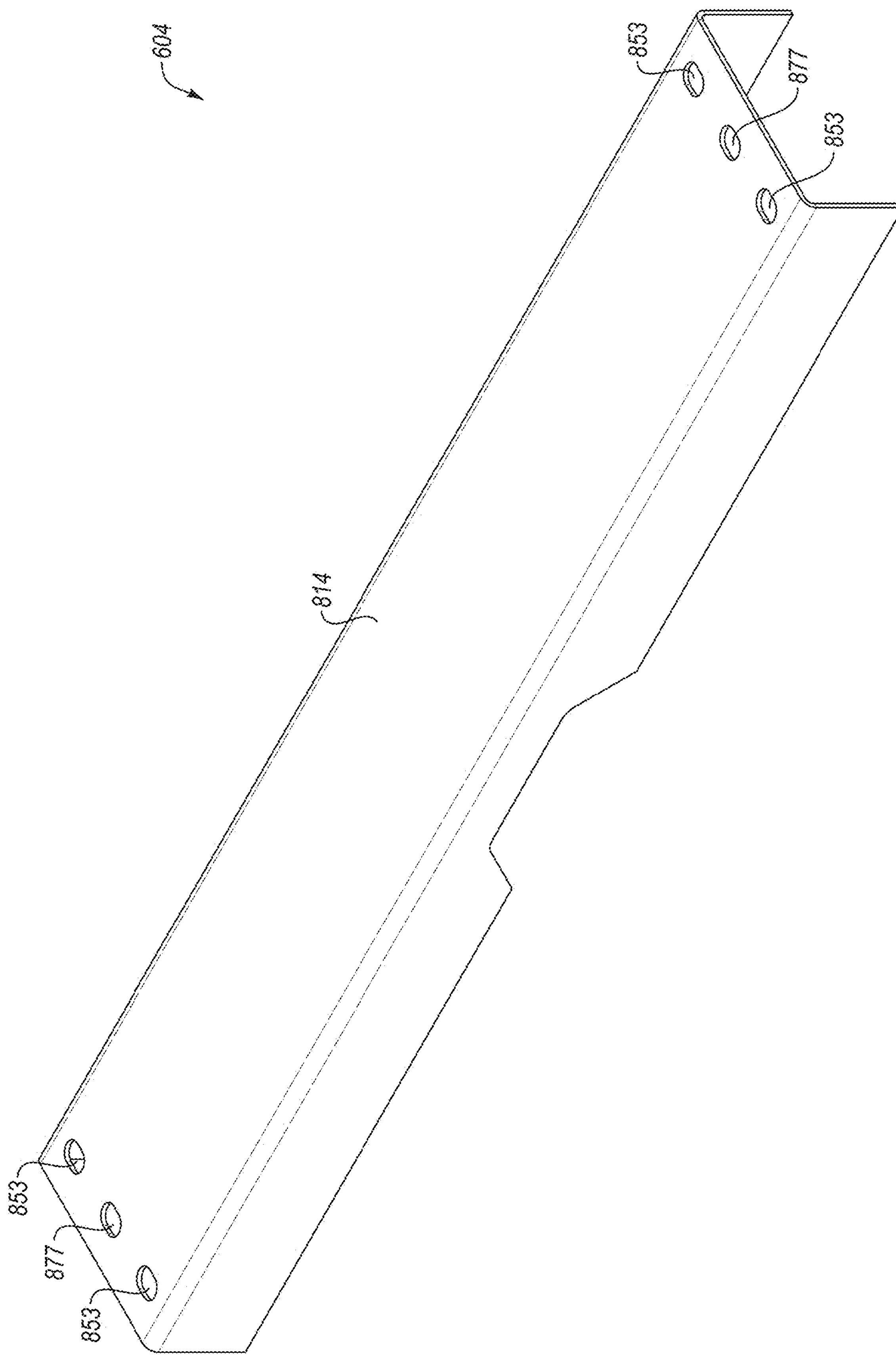


FIG. 8

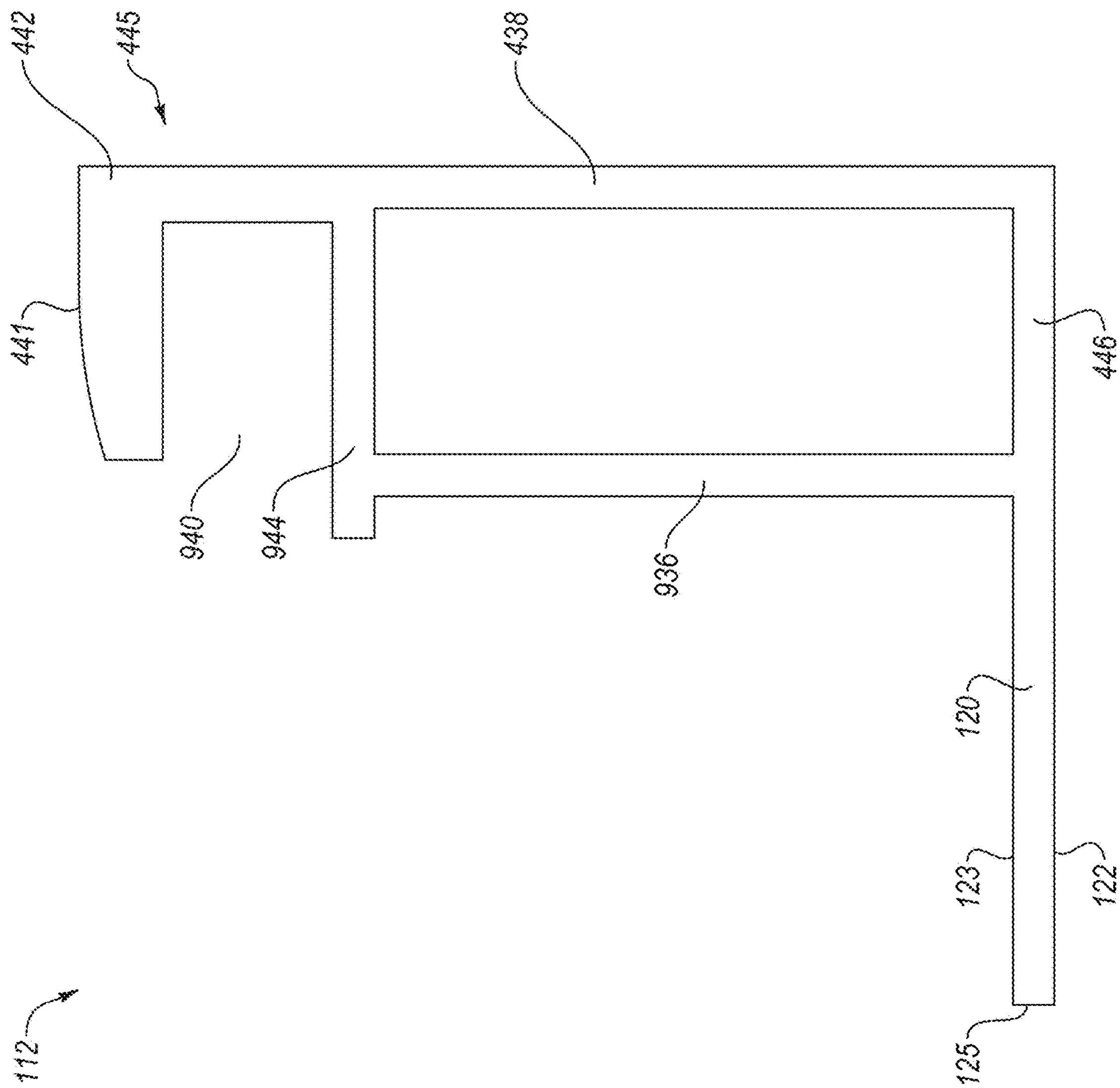
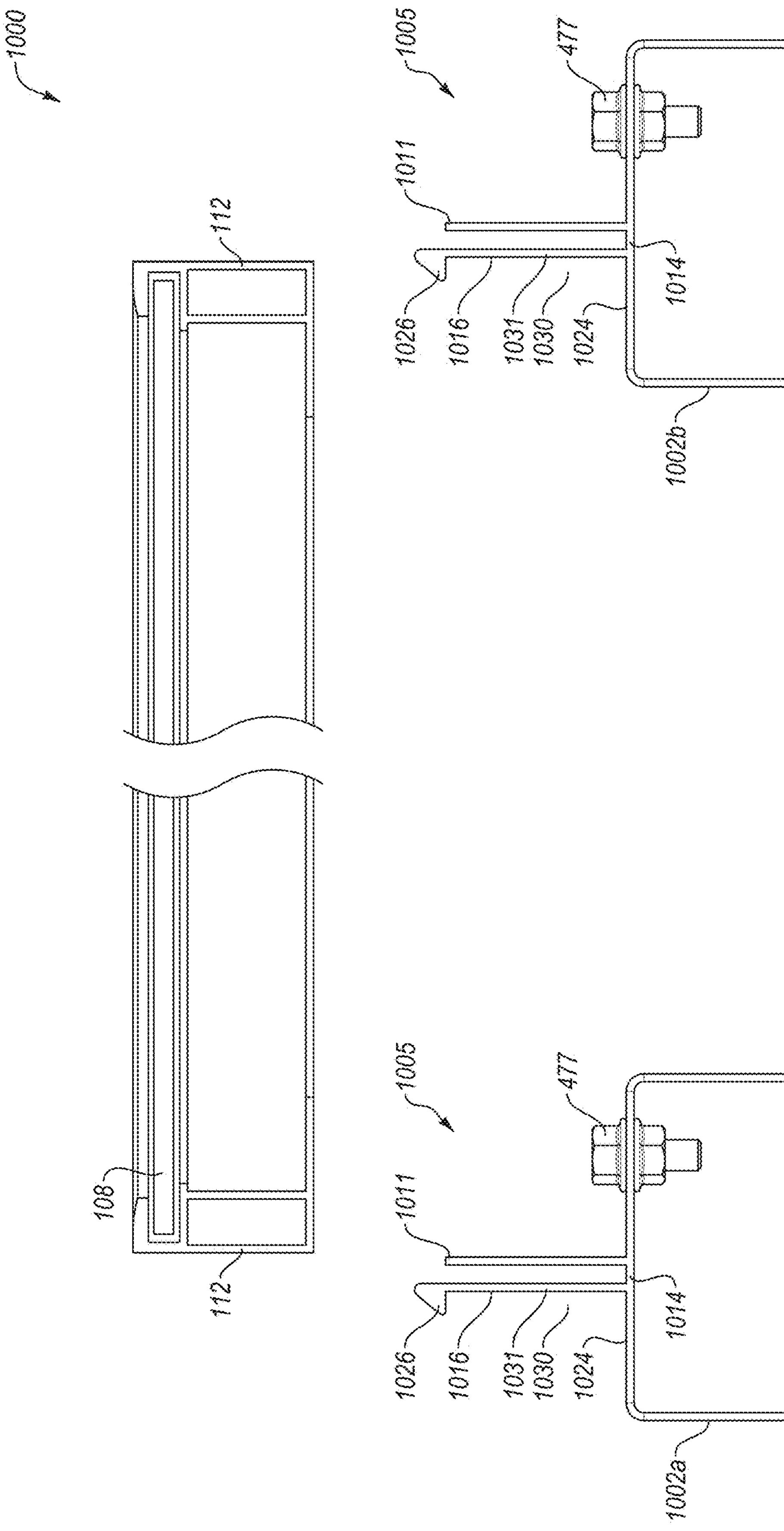


FIG. 9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1024
1002a

FIG. 10

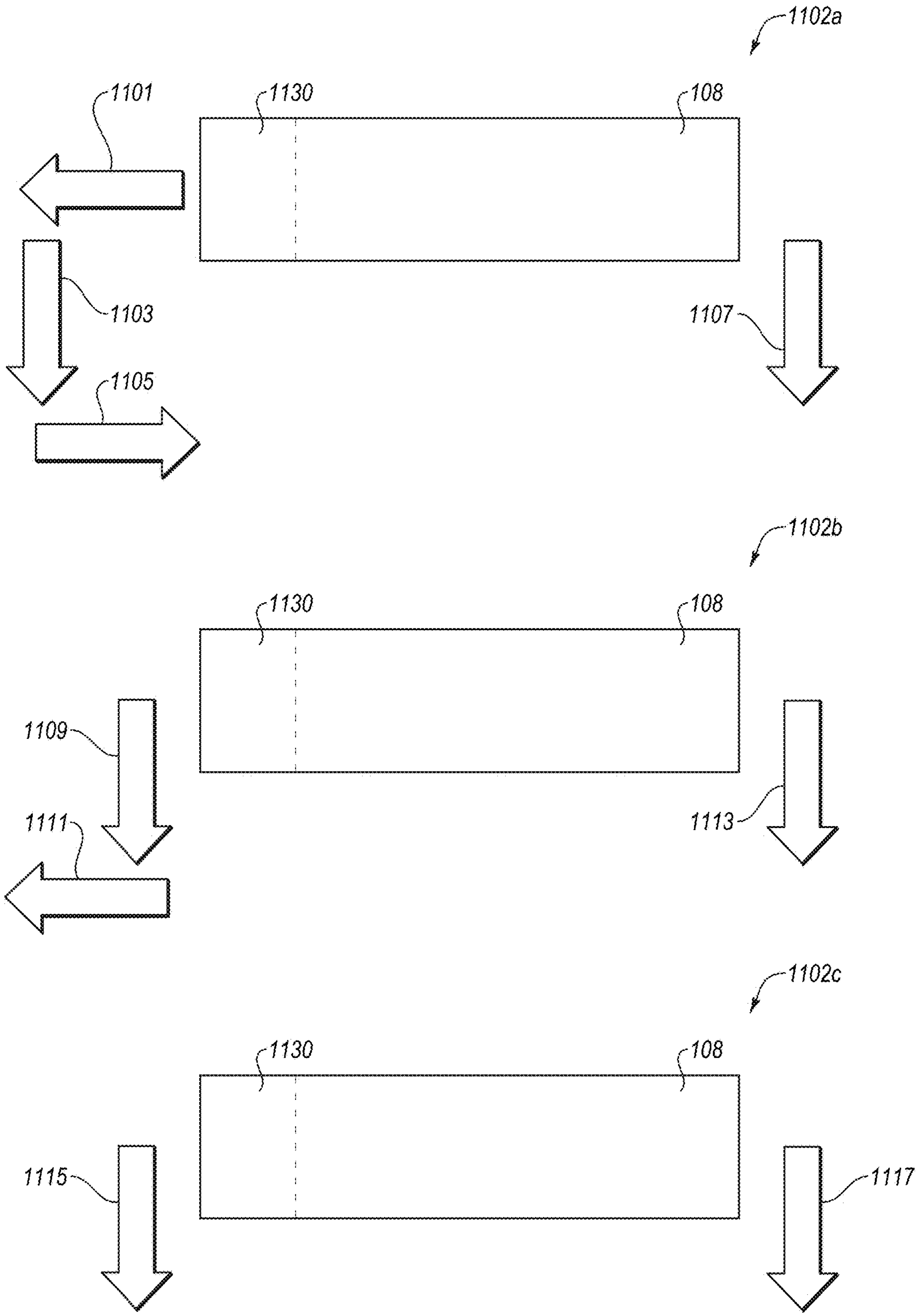
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FIG. 11

MOUNTING RAILS

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This patent application claims the benefit of and priority to U.S. Provisional App. No. 63/563,183 filed Mar. 8, 2024, titled "MOUNTING RAILS," which is incorporated in the present disclosure by reference in its entirety.

FIELD

[0002] The embodiments discussed in the present disclosure are related to solar installations and, more particularly, to mounting rails for use in a solar installation.

BACKGROUND

[0003] Solar installations including solar farms, photovoltaic (PV) plants, solar tracking systems, fixed solar systems, and other PV systems include large numbers of PV modules that collect sunlight and generate energy. During installation, the PV modules may be coupled to corresponding mounting rails to position the PV modules within the solar installations. In particular, the PV modules may be coupled to the corresponding mounting rails via module frames of the PV modules and module mounting systems.

[0004] Some module mounting systems use fasteners such as nuts, bolts, and clips to couple the PV modules to the mounting rails. During installation, installers will pass the fasteners through the module frames and the mounting rails on multiple sides of the PV modules and then secure the fasteners. The installation of the fasteners may require a significant amount of time during the installation of the solar installations. For example, some module mounting systems use four or more bolts per PV module with each fastener needing to be passed through the mounting rail and the module frame and then secured. In addition, some module mounting systems cause the installers to manually position the PV modules and hold the PV modules in place during installation of the PV modules and/or the fasteners, which may require multiple installers. For instance, one or more installers may maintain the position of the PV module while one or more other installers install the fasteners.

[0005] Therefore, there is a need to ease the installation process of coupling the PV modules to the mounting rails and/or permit installation without installers manually positioning and maintaining the position of the PV modules.

[0006] The subject matter claimed in the present disclosure is not limited to embodiments that solve any disadvantages or that operate only in environments such as those described above. Rather, this background is only provided to illustrate one example technology area where some embodiments described in the present disclosure may be practiced.

SUMMARY

[0007] This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential characteristics of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter.

[0008] Exemplary embodiments of the present disclosure address the problems experienced in solar installations, including problems associated with coupling the PV modules to the mounting rails. Disclosed embodiments include one or more mounting rails that are configured to couple to PV modules via a tooled process and/or a toolless process. One embodiment includes a mounting rail that includes a hooked mechanism that couples a first PV module to the mounting rail and an attachment feature that couples a second PV module to the mounting rail. The hooked mechanism at least partially defines an aperture that receives a portion of a first module frame of the first PV module. The hooked mechanism also physically engages with a surface of the first module frame to couple the first module frame to the mounting rail. In this manner, the first the PV module is coupled to the mounting rail via the hooked mechanism (e.g., the toolless process).

[0009] The attachment feature may interface with a second module frame to couple the second PV module to the mounting rail. In some embodiments, the attachment feature may include fasteners and openings defined by the mounting rail. During installation, the openings of the attachment feature may be aligned with openings defined by the second module frame to permit the fasteners pass through the openings to couple the second PV module to the mounting rail. In this manner, the second PV module is coupled to the mounting rail via the attachment feature (e.g., the tooled process).

[0010] Therefore, the mounting rail may couple to the first PV module via a different process than how it couples to the second PV module. The toolless process may reduce a complexity of installing the PV modules and/or reduce an amount of time to install the PV modules. In addition, the omission of the fasteners to couple the first PV module to the mounting rail reduces the cost to install the solar installations.

[0011] One embodiment may include hooked mounting rails and attachment mounting rails that are positioned in an asymmetrical arrangement (e.g., an alternating pattern). The hooked mounting rails and the attachment mounting rails may couple to PV modules in different ways. For example, the hooked mounting rails may include multiple instances of the hooked mechanism to couple corresponding PV modules to the hooked mounting rail (e.g., the toolless process). As another example, the attachment mounting rails may include the attachment feature configured to couple corresponding PV modules to the attachment mounting rail using fasteners (e.g., the tooled process).

[0012] Therefore, the hooked mounting rails and the attachment mounting rails couple to the PV modules via different processes. The toolless process may reduce a complexity of installing the PV modules and/or reduce an amount of time to install the PV modules. In addition, the omission of the fasteners to couple the PV modules to the hooked mounting rails reduces the cost to install the solar installations.

[0013] The object and advantages of the embodiments will be realized and achieved at least by the elements, features, and combinations particularly pointed out in the claims. Both the foregoing summary and the following detailed description are exemplary and explanatory and are not restrictive.

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] Example embodiments will be described and explained with additional specificity and detail through the use of the accompanying drawings in which:

[0015] FIGS. **1**A-**1**C illustrate a front cross-sectional view, a back cross-sectional view, and a top view of a torque tube interface including an example mounting rail and an example mounting clamp to couple example PV modules to a torque tube;

[0016] FIGS. **2**A-**2**G illustrate a perspective view, a first side view, a second side view, a top view, a bottom view, a front view, and a back view of one example of the mounting rail of FIGS. **1**A-**1**C;

[0017] FIG. **3** illustrates a perspective view of one example of a mounting rail coupled to the example module frames;

[0018] FIGS. **4**A-**4**C illustrate a left perspective view, a right perspective view, and a front view of a torque tube interface including an example mounting rail coupling example PV modules to a torque tube;

[0019] FIGS. **5**A-**5**G illustrate a perspective view, a first side view, a second side view, a top view, a bottom view, a front view, and a back view of the example mounting rail of FIGS. **4**A-**4**C;

[0020] FIGS. **6**A and **6**B illustrate a top view and a front view of an example torque tube interface that includes an example hooked mounting rail and an example attachment mounting rail positioned in an asymmetrical arrangement to couple example PV modules to a torque tube;

[0021] FIGS. **7**A-**7**E illustrate a perspective view, a side view, a top view, a bottom view, and a front view of the example hooked mounting rail of FIGS. **6**A and **6**B;

[0022] FIG. **8** illustrates a perspective view of the example attachment mounting rail of FIGS. **6**A and **6**B;

[0023] FIG. **9** illustrates one example of a module frame;

[0024] FIG. **10** illustrates a front view of another torque tube interface including example mounting rails to couple an example PV module to a torque tube; and

[0025] FIG. **11** illustrates graphical representations of sequences of movements to interface an example PV module with different mounting rails described in the present disclosure;

[0026] all according to at least one embodiment described in the present disclosure.

DETAILED DESCRIPTION

[0027] Embodiments of the present disclosure will be explained with reference to the accompanying figures. It is to be understood that the figures are diagrammatic and schematic representations of such example embodiments, and are not limiting, nor are they necessarily drawn to scale. In the figures, features with like numbers indicate like structure and function unless described otherwise.

[0028] Referring to FIGS. **1**A-**1**C, an example of one torque tube interface **100** including an example mounting rail **102** and an example mounting clamp **104** to couple example PV modules **108***a-b* to a torque tube **106** is shown. FIG. **1**A illustrates a front cross-sectional view of the torque tube interface **100** and the torque tube **106** with sidewalls of the PV modules **108***a-b* omitted. FIG. **1**B illustrates a back cross-sectional view of the torque tube interface **100** and the torque tube **106** with sidewalls of the PV modules **108***a-b* omitted. FIG. **1**C illustrates a top view of the torque tube interface **100** and the torque tube **106** with PV panels **110** of the PV modules **108***a-b* omitted. Although the mounting rail **102** is illustrated and described as coupling the PV modules **108***a-b* to the torque tube **106**, it is appreciated that the embodiments described in the present disclosure may be implemented to couple the PV modules **108***a-b* to any appropriate solar component.

[0029] In the example shown, each of the PV modules **108***a-b* includes a PV panel **110**. Each of the PV panels **110** may be coupled to corresponding module frames **112***a-b*. In addition, the module frames **112***a-b* are coupled to the torque tube interface **100** to couple the PV modules **108***a-b* to the torque tube **106**. In particular, the mounting rail **102** is coupled to the mounting clamp **104** and the torque tube **106** and the PV modules **108***a-b* are coupled to the mounting rail **102**.

[0030] In some embodiments, the mounting rail **102** may include an asymmetrical configuration to couple to the PV modules **108***a-b* in different ways. For example, as discussed in more detail below, the mounting rail **102** may couple to the module frame **112***a* of the PV module **108***a* via one or more hooked mechanisms **116***a-d* (each of the hooked mechanisms **116***a-d* are shown in FIG. **1**C), one or more retention members **118** (each of the retention members **118** are shown in FIG. **1**C), or both. As another example, as discussed in more detail below, the mounting rail **102** may couple to the PV module **108***b* via an attachment feature **105**.

[0031] The mounting rail **102** may include an upper portion **114** that is connected to a connecting portion **134**. The upper portion **114** may engage with a bottom surface **122***a* of the module frame **112***a*. In addition, the hooked mechanisms **116***a-d* (generally referred to in the present disclosure as hooked mechanism **116**), the upper portion **114**, or both may at least partially define one or more apertures **130** configured to receive a return flange **120***a* of the module frame **112***a*. The apertures **130** may receive an edge **125** of the return flange **120***a* to position a portion of the return flange **120***a* between the hooked mechanism **116** and the upper portion **114**.

[0032] The apertures **130** may receive the return flange **120***a* to couple the module frame **112***a* to the mounting rail **102**. In addition, the hooked mechanism **116** may physically engage with a top surface **123** of the return flange **120***a* to couple the module frame **112***a* to the mounting rail **102**. The apertures **130** may receive the return flange **120***a* to prevent the bottom surface **122***a* from unintentionally disengaging from the upper portion **114**. Further, the hooked mechanism **116** may engage with the top surface **123** of the return flange **120***a* to maintain the engagement between the upper portion **114** and the bottom surface **122***a*. In other words, the hooked mechanism **116** and the upper portion **114** may sandwich part of the return flange **120***a* between the hooked mechanism **116** and the upper portion **114** to prevent the module frame **112***a* from moving away from a surface **124** of the upper portion **114** (e.g., moving in an upward direction in FIG. **1**A).

[0033] During installation, the PV module **108***a* may be positioned such that the bottom surface **122***a* physically engages with the upper portion **114**. In addition, the PV module **108***a* may be positioned such that the edge **125** of the return flange **120***a* is proximate to the apertures **130**. The PV module **108***a* may be moved such that the module frame **112***a* moves along a width of the mounting rail **102** (e.g., along a left direction in FIG. **1**A) to cause the edge **125** to enter the apertures and the hooked mechanism **116** to engage with the return flange **120***a*. In particular, the hooked mechanism **116** may engage with the top surface **123** to urge the return flange **120***a* towards the upper portion **114**. In addition, the return flange **120***a* may be sandwiched between the hooked mechanism **116** and the upper portion **114**. In some embodiments, the hooked mechanism **116** may prevent the module frame **112***a* from moving too far along the width of the mounting rail **102** (e.g., moving too far left in FIG. **1**A) to prevent the return flange **120***a* from falling off the upper portion **114**. In this manner, the mounting rail **102** couples to the PV module **108***a* via the module frame **112***a* and the hooked mechanism **116** without the use of any fasteners or tools (e.g. a toolless process). Sequences of movements to interface an example PV module with example mounting rails are described in more detail below in relation to FIG. **11**.

[0034] In some embodiments, the mounting rail **102** may include the one or more retention members **118** (generally referred to in the present disclosure as the retention member **118**). In other embodiments, the retention member **118** may be omitted. The retention member **118** may facilitate a position of the module frame **112***a*. In addition, the retention member **118** may support and prevent movement of the module frame **112***a* when coupled to the mounting rail **102**. The retention member **118** may interface with openings **127** (shown in FIG. **1**C) defined by the module frame **112***a* to prevent movement of the module frame **112***a* in one or more directions. For example, the retention member **118**, when interfaced with the openings **127**, may prevent the module frame **112***a* from moving along a length of the mounting rail **102** (e.g., moving into or out of the page of FIG. **1**A), the width of the mounting rail **102** (e.g., moving left or right in FIG. **1**A), or both.

[0035] During installation, the PV module **108***a* may be moved along the width of the mounting rail **102** (e.g., along the left direction in FIG. **1**A) to cause the edge **125** and/or the bottom surface **122***a* to engage with the retention member **118**. The edge **125** engaging with retention member **118** may cause the retention member **118** to selectively deform or move to transition from a first position (e.g., an equilibrium position as shown in FIGS. **1**A-**1**C and FIGS. **2**A-**2**G) to a second position. In other words, the edge **125** engaging with the retention member **118** may cause the retention member to move out of the way of the return flange **120***a* and permit the edge **125** to enter the apertures **130**. The bottom surface **122***a* engaging with the retention member **118** may cause the retention member **118** to remain in the second position.

[0036] When in the second position, the retention member **118** may experience a spring force (e.g., a restoring force) urging the retention member **118** back to the first position. However, the bottom surface **122***a*, when engaged with the retention member **118**, may prevent the retention member **118** from returning to the first position. The PV module **108***a* may be moved along the upper portion **114** until the retention member **118** is aligned with the corresponding opening **127**. When the retention member **118** is aligned with the corresponding opening **127**, the spring force may cause the retention member **118** to return to the first position or partially to the first position and be received by the corresponding opening **127**. In this manner, the retention member **118** may interface with the module frame **112***a* to support and prevent movement of the module frame **112***a*.

[0037] The attachment feature **105** may couple the PV module **108***b* to the mounting rail **102** using fasteners (not shown in FIGS. **1**A-**1**C) (e.g., a tooled process). In some embodiments, the attachment feature **105** may couple the PV module **108***b* to the mounting rail **102** in a way that is different than how the hooked mechanism **116** couples the PV module **108***a* to the mounting rail **102**.

[0038] In some embodiments, the attachment feature **105** may include one or more openings **141** (shown in FIG. **1**C) defined by the upper portion **114**. The openings **141** of the attachment feature **105** may align with openings **129** defined by a return flange **120***b* of the module frame **112***b*. The openings **141** of the attachment feature **105** and the openings **129** of the return flange **120***b* may receive the fasteners of the attachment feature **105** to couple the PV module **108***b* to the mounting rail **102**.

[0039] During installation, the PV module **108***b* may be positioned such that a bottom surface **122***b* of the module frame **112***b* physically engages with the upper portion **114**. In addition, during installation, the PV module **108***b* may be positioned such that the openings **141** of the attachment feature **105** are aligned with the openings **129** of the return flange **120***b*. The openings **141** of the attachment feature **105** and the openings **129** of the return flange **120***b* may receive the fasteners. The fasteners may draw the module frame **112***b* towards the upper portion **114** and couple the module frame **112***b* to the mounting rail **102**. For example, the fasteners may include bolts and nuts, which interface with each other to draw the module frame **112***b* towards the upper portion **114**. In this manner, the mounting rail **102** couples to the PV module **108***b* via the module frame **112***b* and the fasteners. In some embodiments, the fasteners of the attachment feature **105** may include bolts, nuts, threaded fasteners, thru-bolts, clips, top-down clamps, rivets, or some combination thereof.

[0040] With further reference to FIGS. **1**A-**1**C and FIGS. **2**A-**2**G, the upper portion **114** includes the surface **124** that physically engages with the bottom surface **122***a* of the module frame **112***a* and the bottom surface **122***b* of the module frame **112***b*. In some embodiments, the surface **124** may include a first part **235** (shown in FIGS. **2**A and **2**D) and a second part **237** (shown in FIGS. **2**A and **2**D). The first part **235** may physically engage with the bottom surface **122***a* of the module frame **112***a*. The second part **237** may physically engage with the bottom surface **122***b* of the module frame **112***b*.

[0041] As shown, the hooked mechanism **116** overlaps a portion of the surface **124** to at least partially define the apertures **130**. In particular, each of the hooked mechanisms **116***a-d* may include a first curved portion **231***a-d* (generally referred to in the present disclosure as the first curved portion **231**), a second curved portion **233***a-d* (generally referred to in the present disclosure as the second curved portion **233**), or both that at least partially overlap the surface **124**. The first curved portion **231** may extend from an edge **226** of the upper portion **114** and curve back toward the upper portion **114** such that the first curved portion **231** includes a first radius of curvature. The second curved portion **233** may extend from the first curved portion **231** away from the surface **124** such that the second curved portion **233** includes a second radius of curvature. The first curved portion **231**, the second curved portion **233**, or both may define the apertures **130**. The second curved portion **233** may extend in a direction away from the surface **124** such that a size of the apertures **130** varies to accommodate a variety of heights of the return flange **120***a*.

[0042] The apertures **130** may be defined to receive different portions of the edge **125** to couple the module frame

112*a* to the mounting rail **102** as discussed above. The hooked mechanism **116**, when engaged with the return flange **120***a*, may apply a force on the top surface **123** to urge the return flange **120***a* towards the upper portion **114**. For example, the return flange **120***a* entering the apertures **130** may cause the hooked mechanism **116** to selectively deform or move to cause the hooked mechanism **116** to apply the force on the top surface **123**. In this way, the hooked mechanism **116** may prevent the bottom surface **122***a* from unintentionally disengaging from the upper portion **114**.

[0043] In the example shown, the hooked mechanism **116** includes multiple hooked mechanisms **116***a-d* connected to the edge **226** of the upper portion **114** and positioned at different locations along the edge **226**. The hooked mechanisms **116***a-d* may be positioned at the different locations to interface with different portions of the return flange **120***a*. In addition, in the example shown, the hooked mechanism **116** includes the hooked mechanism **116***a* positioned at a longitudinal end **228***a* of the upper portion **114** and the hooked mechanism **116***d* positioned at a longitudinal end **228***b* of the upper portion **114**. Alternatively, one or more of the hooked mechanisms **116***a* or **116***d* may be positioned a distance from the longitudinal ends **228***a-b*. For example, the hooked mechanism **116***a* may be a distance from the longitudinal end **228***a*, the hooked mechanism **116***d* may be a distance from the longitudinal end **228***b*, or both.

[0044] The hooked mechanism **116** is illustrated in FIGS. **1**A-**2**G as including four hooked mechanisms **116***a-d* for example purposes. It is understood that the hooked mechanism **116** may include any appropriate number of hooked mechanisms. For example, the hooked mechanism **116** may include one, two, or more hooked mechanisms.

[0045] The retention member **118** may include a portion that extends in a direction away from the surface **124** such that the portion of the retention member **118** is positioned above the surface **124** when in the first position. In some embodiments, the retention member **118** may include one or more bent pieces of material that are configured to interface with the module frame **112***a* to maintain a position of the module frame **112***a* relative to the upper portion **114**. The retention member **118** may be connected to the upper portion **114** to create a spring tension such that transitioning from the first position to the second position causes the retention member **118** to experience the spring force.

[0046] The retention member **118** may engage with the edge **125** and/or the bottom surface **122***a* such that an end of the retention member **118** is flush with the surface **124** when in the second position. Additionally or alternatively, at least a portion of the retention member **118** may be disposed in an opening defined by the upper portion to permit the edge **125** to be received by the apertures **130**.

[0047] In the example shown, the retention member **118** includes two retention members **118** connected to the upper portion **114**. However, it is understood that the retention member **118** may include any appropriate number of retention members. For example, the retention member **118** may include one, two, three or more retention members. The retention member **118** is illustrated as including two bent clips for example purposes. It is understood that the retention member **118** may include any appropriate device. For example, the retention member **118** may include a tab, a clip, a stud, a spring-loaded tab, a spring-loaded clip, a spring-loaded tab, or any other appropriate device. In addition, each of the retention members **118** are illustrated as being positioned between two hooked mechanisms **116***a-d* for example purposes.

[0048] In some embodiments, the mounting rail **102** may include one or more alignment members **232** (generally referred to in the present disclosure as the alignment member **232**) that extend away from the surface **124**. The alignment member **232** may physically engage with sidewalls (such as denoted **438** in FIG. **9**) of the module frames **112***a-b* to align the PV modules **108***a-b* relative to the upper portion **114**. In addition, the alignment member **232** may be sized to prevent the PV modules **108***a-b* from contacting each other. Further, the alignment member **232** may physically engage with the sides of the module frames **112***a-b* to prevent the module frames **112***a-b* from moving too far along the width of the mounting rail **102**.

[0049] As discussed above, the attachment feature **105** may include the openings **141** to receive the fasteners to couple the module frame **112***b* to the mounting rail **102**. In some embodiments, the fasteners may include bolts that traverse the openings **141** of the attachment feature **105** to interface with the module frame **112***b* and/or nuts (not illustrated) to couple the module frame **112***b* to the mounting rail **102**. For example, the bolts may traverse the openings **141** of the attachment feature **105** and the nuts may thread onto ends of the bolts. Heads of the bolts may be oversized compared to the openings **141** defined by the surface **124** to prevent the bolts from passing through the openings defined by the surface **124** and/or the openings **129**. In some embodiments, the openings **141** defined by the surface **124** may include threaded portions configured to interface with threaded portions of the bolts.

[0050] In some embodiments, the mounting rail **102** may include a single unitary piece of material in which the hooked mechanism **116**, the retention member **118**, the alignment member **232**, or some combination thereof are punched or otherwise formed from. In other embodiments, the mounting rail **102** may include multiple pieces of material. The hooked mechanism **116** may be connected to the mounting rail **102** via a weld, a rivet, a threaded fastener, an adhesive, a clinch joint, or a snap-in feature. In these and other embodiments, the mounting rail **102** may include aluminum, steel, or any other appropriate material. In some embodiments, the mounting rail **102** may be formed using an extrusion process.

[0051] Although the mounting rail **102** is illustrated as including the hooked mechanism **116** and the retention member **118** only on one side of the mounting rail **102** (e.g., along the edge **226**), the mounting rail **102** may include additional hooked mechanisms, retention members, or both along the opposite side to couple the module frame **112***b* to the mounting rail **102**.

[0052] With reference to FIG. **3**, an example mounting rail **302** coupled to the module frames **112***a-b* is shown. The mounting rail **302** may couple the module frames **112***a-b* to a torque tube, a mounting clip, or any other appropriate solar component.

[0053] In some embodiments, the mounting rail **302** may include an asymmetrical configuration to couple the mounting rail **302** to the module frames **112***a-b* in different ways. For example, as discussed in more detail below, the mounting rail **302** may couple to the module frame **112***a* via an elongated hooked mechanism **316** (generally referred to in the present disclosure as the hooked mechanism **316**), the one or more retention members **118**, or both. As another example, as discussed in more detail below, the mounting rail **302** may couple to the module frame **112***b* via an attachment feature (not shown in FIG. **3**).

[0054] The mounting rail **302** may include an upper portion (not shown in FIG. **3**) connected to a connecting portion **346**. The upper portion may engage with the bottom surface **122***a* of the module frame **112***a*. In addition, the hooked mechanism **316**, the upper portion, or both may at least partially define an aperture (not shown in FIG. **3**) configured to receive the return flange **120***a* of the module frame **112***a*. The aperture may receive the edge **125** of the return flange **120***a* to position a portion of the return flange **120***a* between the hooked mechanism **316** and the upper portion. The aperture of the mounting rail **302** may operate the same or similar to the apertures **130** of FIGS. **1**A-**2**G.

[0055] The aperture may receive the return flange **120***a* to couple the module frame **112***a* to the mounting rail **302**. In addition, the hooked mechanism **316** may physically engage with the top surface **123** of the return flange **120***a* to couple the module frame **112***a* to the mounting rail **302**. The aperture may receive the return flange **120***a* to prevent the bottom surface **122***a* from unintentionally disengaging from the upper portion. Further, the hooked mechanism **316** may engage with the top surface **123** of the return flange **120***a* to maintain the engagement between the upper portion and the bottom surface **122***a*. In other words, the hooked mechanism **316** and the upper portion may sandwich part of the return flange **120***a* between the hooked mechanism **316** and the upper portion to prevent the module frame **112***a* from moving away from a surface of the upper portion (e.g., moving in an upward direction in FIG. **3**).

[0056] During installation, the module frame **112***a* may be positioned such that the bottom surface **122***a* physically engages with the upper portion of the mounting rail **302**. In addition, the module frame **112***a* may be positioned such that the edge **125** of the return flange **120***a* is proximate to the aperture. The module frame **112***a* may be moved such that the module frame **112***a* moves along a width of the mounting rail **302** to cause the edge **125** to enter the aperture and the hooked mechanism **316** to engage with the return flange **120***a*. In particular, the hooked mechanism **316** may engage with the top surface **123** to urge the return flange **120***a* towards the upper portion. In addition, the return flange **120***a* may be sandwiched between the hooked mechanism **316** and the upper portion of the mounting rail **302**. In some embodiments, the hooked mechanism **316** may prevent the module frame **112***a* from moving too far along the width of the mounting rail **302** to prevent the return flange **120***a* from falling off the upper portion. In this manner, the mounting rail **302** couples to the module frame **112***a* and the hooked mechanism **316** without the use of any fasteners or tools (e.g., the toolless process). Sequences of movements to interface an example PV module with example mounting rails are described in more detail below in relation to FIG. **11**.

[0057] The attachment feature may couple the PV module **108***b* to the mounting rail **302** using fasteners (not shown in FIG. **4**) (e.g., the tooled process). In some embodiments, the attachment feature may couple the PV module **108***b* to the mounting rail **302** in a way that is different than how the hooked mechanism **316** couples the PV module **108***a* to the mounting rail **302**. The attachment feature of the mounting rail **302** may operate the same as or similar to the attachment feature **105** discussed above in relation to FIGS. **1**A-**2**G.

[0058] In some embodiments, the attachment feature may include one or more openings (not shown in FIG. **4**) defined by the upper portion. The openings of the attachment feature may align with the openings **129** defined by the module frame **112***b*. The openings of the attachment feature and the openings **129** may receive the fasteners of the attachment feature to couple the module frame **112***b* to the mounting rail **402**.

[0059] The surface of the upper portion may physically engage with the bottom surface **122***a* of the module frame **112***a* and the bottom surface **122***b* of the module frame **112***b*. In some embodiments, the surface of the upper portion may include a first part (not shown in FIG. **3**) and a second part (not shown in FIG. **3**). The first part may physically engage with the bottom surface **122***a* of the module frame **112***a*. The second part may physically engage with the bottom surface **122***b* of the module frame **112***b*.

[0060] The hooked mechanism **316** overlaps a portion of the surface of the upper portion to at least partially define the aperture. In particular, the hooked mechanism **316** may include a first curved portion **331**, a second curved portion **333**, or both that at least partially overlap the surface of the upper portion. The first curved portion **331** may extend from an edge (not shown in FIG. **3**) of the upper portion and curve back toward the upper portion. The second curved portion **333** may extend from the first curved portion **331** away from the surface of the upper portion. The first curved portion **331**, the second curved portion **333**, or both may define the aperture. The second curved portion **333** may extend in a direction away from the surface of the upper portion such that a size of the aperture varies to accommodate a variety of heights of the return flange **120***a*.

[0061] The hooked mechanism **316**, when engaged with the return flange **120***a*, may apply a force on the top surface **123** to urge the return flange **120***a* towards the upper portion. For example, the return flange **120***a* entering the aperture may cause the hooked mechanism **316** to selectively deform or move to cause the hooked mechanism **316** to apply the force on the top surface **123**. In this way, the hooked mechanism **316** may prevent the bottom surface **122***a* from unintentionally disengaging from the upper portion of the mounting rail **302**.

[0062] In the example shown, the hooked mechanism **316** includes a single elongated hooked mechanism connected to the edge of the upper portion and positioned at least proximate a center of the edge of the upper portion. The hooked mechanism **316** may extend continuously along at least part of a length of the edge of the upper portion. In the example shown, the hooked mechanism **316** extends along a part of a distance between the retention member **118**.

[0063] In some embodiments, the mounting rail **302** may include the retention member **118**. In other embodiments, the retention member **118** may be omitted. Additionally or alternatively, the mounting rail **302** may include the alignment member **232** (not shown in FIG. **3**).

[0064] In some embodiments, the mounting rail **302** may include a single unitary piece of material in which the hooked mechanism **316**, the retention member **118**, or both are punched or otherwise formed from. In other embodiments, the mounting rail **302** may include multiple pieces of material. The hooked mechanism **316** may be connected to the mounting rail **302** via a weld, a rivet, a threaded fastener, an adhesive, a clinch joint, or a snap-in feature. In these and other embodiments, the mounting rail **302** may include aluminum, steel, or any other appropriate material. In some embodiments, the mounting rail **302** may be formed using an extrusion process.

[0065] Although the mounting rail **302** is described as including the hooked mechanism **316** and the retention member **118** only on one side of the mounting rail **302**, the mounting rail **302** may include additional hooked mechanisms, retention members **118**, or both along the opposite side to couple the module frame **112***b* to the mounting rail **302**.

[0066] Referring to FIGS. **4**A-**4**C, an example torque tube interface **400** including an example mounting rail **402** coupling the PV modules **108***a-b* to the torque tube **106** is shown. FIG. **4**A illustrates a left perspective view of the mounting rail **402**, the PV modules **108***a-b*, and the torque tube **106** with the PV panels **110** of the PV modules **108***a-b* omitted. FIG. **4**B illustrates a right perspective view of the mounting rail **402**, the PV modules **108***a-b*, and the torque tube **106** with the PV panels **110** of the PV modules **108***a-b* omitted. FIG. **4**C illustrates a front view of the mounting rail **402**, the PV modules **108***a-b*, and the torque tube **106** with sidewalls of the PV modules **108***a-b* omitted. Although the mounting rail **402** is illustrated and described as coupling the PV modules **108***a-b* to the torque tube **106**, it is appreciated that the embodiments described in the present disclosure may be implemented to couple the PV modules **108***a-b* to any appropriate solar component.

[0067] In some embodiments, the mounting rail **402** may include an asymmetrical configuration to couple to the PV modules **108***a-b* in different ways. For example, as discussed in more detail below, the mounting rail **402** may couple to the module frame **112***a* of the PV module **108***a* via one or more hooked mechanisms **416***a-b* (generally referred to in the present disclosure as hooked mechanism **416**). As another example, as discussed in more detail below, the mounting rail **402** may couple to the module frame **112***b* of the PV module **108***b* via an attachment feature **405**.

[0068] The mounting rail **402** may include an upper portion **414**. The upper portion **414** may engage with the bottom surface **122***a* of the module frame **112***a*. In addition, the hooked mechanism **416** may at least partially define one or more apertures **430** configured to receive a portion **445** of the module frame **112***a*. As shown, the portion **445** of the module frame **112***a* includes a sidewall **438**, at least part of a panel portion **442**, at least part of a base portion **446**, or some combination thereof of the module frame **112***a*. The apertures **430** may receive the portion **445** of the module frame **112***a* to position at least part of the portion **445** of the module frame **112***a* between the hooked mechanism **416** and the upper portion **414**.

[0069] The apertures **430** may receive the portion **445** of the module frame **112***a* to couple the module frame **112***a* to the mounting rail **402**. In addition, the hooked mechanism **416** may physically engage with a top surface **441** of the panel portion **442** to couple the module frame **112***a* to the mounting rail **402**. The apertures **430** may receive the portion **445** of the module frame **112***a* to prevent the bottom surface **122***a* from unintentionally disengaging from the upper portion **414**. Further, the hooked mechanism **416** may engage with the top surface **441** of the panel portion **442** to maintain the engagement between the upper portion **414** and the bottom surface **122***a*. In other words, the hooked mechanism **416** and the upper portion **414** may sandwich part of the portion **445** of the module frame **112***a* between the hooked mechanism **416** and the upper portion **414** to prevent the module frame **112***a* from moving away from a surface **424** of the upper portion **414** (e.g., moving in an upward direction in FIG. **4**C).

[0070] During installation, the PV module **108***a* may be positioned such that the bottom surface **122***a* physically engages with the upper portion **414**. In addition, the PV module **108***a* may be positioned such that the portion **445** of the module frame **112***a* is proximate to the apertures **430**. The PV module **108***a* may be moved such that the module frame **112***a* moves along a width of the mounting rail **402** (e.g., along a right direction in FIG. **4**C) to cause the portion **445** of the module frame **112***a* to enter the apertures **430** and the hooked mechanism **416** to engage with the portion **445** of the module frame **112***a*. In particular, the hooked mechanism **416** may engage with the top surface **441** of the panel portion **442** to urge the module frame **112***a* towards the upper portion **414**. In some embodiments, the hooked mechanism **416** may prevent the module frame **112***a* from moving too far along the width of the mounting rail **402** (e.g., moving too far right in FIG. **4**C) to facilitate alignment of the PV module **108***a*. In this manner, the mounting rail **402** couples to the PV module **108***a* via the module frame **112***a* and the hooked mechanism **416** without the use of any fasteners or tools. (e.g. the toolless process). Sequences of movements to interface an example PV module with example mounting rails are described in more detail below in relation to FIG. **11**.

[0071] As shown in FIGS. **4**A-**4**C, the attachment feature **405** couples the PV module **108***b* to the mounting rail **402** using fasteners **471** (shown in FIGS. **4**B and **4**C) (e.g., the tooled process). As shown the fasteners **471** include bolts **477** and nuts **473**. Additionally or alternatively, the fasteners **471** may include rivets, swage fasteners, or any other appropriate tooled attachment feature. In other words, the attachment feature **405** may couple the PV module **108***b* to the mounting rail **402** in a way that is different than how the hooked mechanism **416** couples the PV module **108***a* to the mounting rail **402**.

[0072] With further reference to FIGS. **4**A-**4**C and FIGS. **5**A-**5**G, in some embodiments, the attachment feature **405** may include one or more openings **553** (shown in FIGS. **5**A, **5**D, and **5**E) defined by the upper portion **414**. The openings **553** may align with the openings **129** defined by the return flange **120***b* of the module frame **112***b*. The openings **553** of the attachment feature **405** and the openings **129** defined by the return flange **120***b* (examples of which are shown in FIG. **4**A in relation to the return flange **120** of the module frame **112***a*) may receive the fasteners **471** to couple the PV module **108***b* to the mounting rail **402**.

[0073] During installation, the PV module **108***b* may be positioned such that the bottom surface **122***b* of the module frame **112***b* physically engages with the upper portion **414**. In addition, during installation, the PV module **108***b* may be positioned such that the openings **553** of the attachment feature **405** are aligned with the openings **129** defined by the return flange **120***b*. The openings **553** of the attachment feature **405** and the openings **129** defined by the return flange **120***b* may receive the fasteners **471**. The fasteners **471** may draw the module frame **112***b* towards the upper portion **414** and couple the module frame **112***b* to the mounting rail

402. For example, the bolts **477** may interface with the nuts **473** to draw the module frame **112***b* towards the upper portion **414**.

[0074] In this manner, the mounting rail **402** couples to the PV module **108***b* via the module frame **112***b* and the fasteners **471**. The attachment feature **405** is illustrated and described as including the fasteners **471** and the openings **553** for example purposes. The attachment feature **405** may include any appropriate device to couple the PV module **108***b* to the mounting rail **402**. For example, the attachment feature **405** may include bolts, nuts, threaded fasteners, thru-bolts, clips, top-down clamps, rivets, or some combination thereof.

[0075] The upper portion **414** includes the surface **424**, that physically engages with the bottom surface **122***a* of the module frame **112***a* and the bottom surface **122***b* of the module frame **112***b*. In some embodiments, the surface **424** may include a first part **535** (shown in FIGS. **5**A and **5**D) and a second part **537** (shown in FIGS. **5**A and **5**D). The first part **535** may physically engage with the bottom surface **122***a* of the module frame **112***a*. The second part **537** may physically engage with the bottom surface **122***b* of the module frame **112***b*.

[0076] Each of the hooked mechanisms **416***a-b* may include a first component **531***a-b* (generally referred to in the present disclosure as the first component **531**), a second component **533***a-b* (generally referred to in the present disclosure as the second component **533**), or both. The first component **531** may extend from the surface **424** of the upper portion **414**. The second component **533** may extend from the first component **531** and towards the upper portion **414** such that the second component **533** may include a radius of curvature. The first component **531**, the second component **533**, or both may define the apertures **430**. The second component **533** may extend in a direction towards the upper portion **414** to apply a force on the top surface **441** of the panel portion **442** when engaged with the module frame **112***a*.

[0077] The apertures **430** may be defined to receive different sections of the module frame **112***a* to couple the module frame **112***a* to the mounting rail **402** as discussed above. The hooked mechanism **416**, when engaged with the panel portion **442**, may apply a force on the top surface **441** of the panel portion **442** to urge the module frame **112***a* towards the upper portion **414**. For example, the portion **445** of the module frame **112***a* entering the apertures **430** may cause the hooked mechanism **416** to selectively deform or move to cause the hooked mechanism **416** to apply the force on the top surface **441** of the panel portion **442**. In this way, the hooked mechanism **416** may prevent the bottom surface **122***a* from unintentionally disengaging from the upper portion **414**.

[0078] In the example shown, the hooked mechanism **416** includes multiple hooked mechanisms **416***a-b* connected to the upper portion **414** and positioned at different locations on the upper portion **414**. The hooked mechanisms **416***a-b* may be positioned at the different locations to interface with different sections of the portion **445** of the module frame **112***a*. In addition, in the example shown, the hooked mechanism **416** includes the hooked mechanism **416***a* positioned proximate a longitudinal end **528***a* of the upper portion **414** and the hooked mechanism **416***b* positioned proximate a longitudinal end **528***b* of the upper portion **414**. Alternatively, one or more of the hooked mechanisms **416***a* or **416***b* may be positioned at the longitudinal ends **528***a-b*. For example, the hooked mechanism **416***a* may be at the longitudinal end **528***a*, the hooked mechanism **416***b* may be at the longitudinal end **528***b*, or both.

[0079] The hooked mechanism **416** is illustrated in FIGS. **4**A-**5**G as including two hooked mechanisms **416***a-b* for example purposes. It is understood that the hooked mechanism **416** may include any appropriate number of hooked mechanisms. For example, the hooked mechanism **416** may include one, three, four, or more hooked mechanisms.

[0080] In some embodiments, the hooked mechanism **416** may align the PV module **108***a* relative to the upper portion **414**. In addition, the hooked mechanism **416** may be sized to prevent the PV modules **108***a-b* from contacting each other. Further, the hooked mechanism **416** may physically engage with the portion **445** of the module frame **112***a* to prevent the module frame **112***a* from moving too far along the width of the mounting rail **402**.

[0081] In some embodiments, the fasteners **471** may traverse the openings **553** to interface with the module frame **112***b* and/or the nuts **473** to couple the module frame **112***b* to the mounting rail **402**. For example, the bolts **477** may traverse the openings **553** and the nuts **473** may thread onto ends of the bolts **477**. Heads of the bolts **477** may be oversized compared to the openings **553** to prevent the bolts **477** from passing through the openings **553** and/or the openings **129** defined by the return flange **120***b*. In some embodiments, the openings **553** may include threaded portions configured to interface with threaded portions of the bolts **477**.

[0082] In some embodiments, the mounting rail **402** may include one or more stopper devices (not shown in FIGS. **4**A-**5**G). The stopper devices may be connected to and extend from the upper portion **414**. In some embodiments, the stopper devices may be positioned proximate to and extend generally parallel to the hooked mechanism **416**. The stopper devices may interface with the hooked mechanism **416** during installation of the PV module **108***a* to prevent the hooked mechanism **416** from selectively deforming beyond a pre-determined threshold. In some embodiments, the pre-determined threshold may be equal to or greater than eight degrees from an equilibrium position of the hooked mechanism **416**. The hooked mechanism **416** is illustrated in FIGS. **4**A-**5**G in the equilibrium position.

[0083] In some embodiments, the mounting rail **402** may include the retention member **118**. In other embodiments, the retention member **118** may be omitted.

[0084] In some embodiments, the mounting rail **402** may include a single unitary piece of material in which the hooked mechanism **416** is punched or otherwise formed from. For example, the hooked mechanism **416** may be punched or otherwise formed from the unitary piece of material such that one or more openings **555** are formed. In other embodiments, the mounting rail **402** may include multiple pieces of material. The hooked mechanism **416** may be connected to the mounting rail **402** via a weld, a rivet, a threaded fastener, an adhesive, a clinch joint, or a snap-in feature. In these and other embodiments, the mounting rail **402** may include aluminum, steel, or any other appropriate material. In some embodiments, the mounting rail **402** may be formed using an extrusion process.

[0085] Referring to FIGS. **6**A and **6**B, an example torque tube interface **600** that includes an example hooked mounting rail **602** and an example attachment mounting rail **604**

positioned in an asymmetrical arrangement to couple PV modules **108***a-c* to the torque tube **106** is shown. FIG. **6**A illustrates a top view of the torque tube interface **600**, the PV modules **108***a-c*, and the torque tube **106** with the PV panels **110** of the PV modules **108***a-c* omitted. FIG. **6**B illustrates a front view of the torque tube interface **600**, the PV modules **108***a-c*, and the torque tube **106** with sidewalls of the PV modules **108***a-c* omitted. Although the torque tube interface **600** is illustrated and described as coupling the PV modules **108***a-c* to the torque tube **106**, it is appreciated that the embodiments described in the present disclosure may be implemented to couple the PV modules **108***a-c* to any appropriate solar component.

[0086] In some embodiments, the hooked mounting rail **602** and the attachment mounting rail **604** may couple to the PV modules **108***a-c* in different ways. For example, as discussed in more detail below, the hooked mounting rail **602** may couple to the PV modules **108***a-b* via hooked mechanisms **616***a-b*. As another example, as discussed in more detail below, the attachment mounting rail **604** may couple to the PV modules **108***b-c* via an attachment feature **605**.

[0087] The torque tube interface **600** may include multiple instances of the hooked mounting rail **602** and multiple instances of the attachment mounting rail **604** that are alternatingly positioned on the torque tube **106** in the asymmetrical arrangement. As shown in FIGS. **6**A and **6**B, the hooked mounting rail **602** interfaces with the torque tube **106** at a first location **601** and the attachment mounting rail **604** interfaces with the torque tube **106** at a second location **603**. In addition, an additional attachment mounting rail (not shown in FIGS. **6**A and **6**B) may interface with the torque tube **106** at a third location (not shown in FIGS. **6**A and **6**B) and an additional hooked mounting rail (not shown in FIGS. **6**A and **6**B) may interface with the torque tube **106** at a fourth location (not shown in FIGS. **6**A and **6**B). In some embodiments, the third location may be located on an opposite side of the first location **601** (e.g., the hooked mounting rail **602**) as the second location **603** (e.g., the attachment mounting rail **604**). In these and other embodiments, the fourth location may be located on an opposite side of the second location **603** as the first location **601**.

[0088] A distance between the locations that the hooked mounting rail **602** and the attachment mounting rail **604** interface with the torque tube **106** may correspond to a width of the PV modules **108***a-c*. In other words, a distance between the hooked mounting rail **602** and the attachment mounting rail **604** may be based on the width of the PV modules **108***a-c*. For example, the distance between the first location **601** and the second location **603** may be based on the width of the PV module **108***b*. As another example, the distance between the first location **601** and the third location may be based on the width of the PV module **108***a* and the distance between the second location **603** and the fourth location may be based on the width of the PV module **108***c*.

[0089] As shown in FIGS. **6**A and **6**B, the hooked mounting rail **602** is coupled to a left side of the PV module **108***b* and the attachment mounting rail **604** is coupled to a right side of the PV module **108***b*. In addition, the hooked mounting rail **602** is attached to a right side of the PV module **108***a* and the additional attachment mounting rail is coupled to a left side (not shown in FIGS. **6**A and **6**B) of the PV module **108***a*. Further, the attachment mounting rail **604** is attached to a left side of the PV module **108***c* and the additional hooked mounting rail is coupled to a right side (not shown in FIGS. **6**A and **6**B) of the PV module **108***c*.

[0090] With combined reference to FIGS. **6**A-**7**E, the hooked mounting rail **602** includes hooked mechanisms **616***a-b* and an upper portion **614**. The upper portion **614** may engage with the bottom surfaces **122***a-b* of the module frames **112***a-b*. In addition, the hooked mechanisms **616***a-b* may overlap portions of a surface **624** of the upper portion **614** to at least partially define apertures **630***a-b*. The apertures **630***a-b* may be defined to receive the portions **445** of the corresponding module frames **112***a-b*. The apertures **630***a-b* may receive the portions **445** of the corresponding module frames **112***a-b* to position at least part of the portions **445** between the corresponding hooked mechanisms **616***a-b* and the upper portion **614**.

[0091] The apertures **630***a-b* may receive the portions **445** of the module frames **112***a-b* to couple the corresponding module frames **112***a-b* to the hooked mounting rail **602**. In addition, the hooked mechanisms **616***a-b* may physically engage with the top surfaces **441** of the panel portions **442** to couple the corresponding module frames **112***a-b* to the hooked mounting rail **602**. The apertures **630***a-b* may receive the portions **445** of the module frames **112***a-b* to prevent the corresponding bottom surfaces **122***a-b* from unintentionally disengaging from the upper portion **614**. Further, the hooked mechanisms **616***a-b* may engage with the top surfaces **441** of the panel portions **442** to maintain the engagement between the upper portion **614** and the corresponding bottom surfaces **122***a-b*. In other words, the hooked mechanisms **616***a-b* and the upper portion **614** may sandwich part of the module frames **112***a-b* between the hooked mechanisms **616***a-b* and the upper portion **614** to prevent the corresponding module frames **112***a-b* from moving away from the surface **624** of the upper portion **614** (e.g., moving in an upward direction in FIG. **6**B).

[0092] During installation of the PV module **108***a*, the PV module **108***a* may be positioned such that the bottom surface **122***a* physically engages with the upper portion **614**. In addition, the PV module **108***a* may be positioned such that the portion **445** of the module frame **112***a* is proximate to the aperture **630***a*. The PV module **108***a* may be moved such that the module frame **112***a* moves along a width of the hooked mounting rail **602** (e.g., along a right direction in FIG. **6**B) to cause the portion **445** of the module frame **112***a* to enter the aperture **630***a* and the hooked mechanism **616***a* to engage with the portion **445** of the module frame **112***a*. In particular, the hooked mechanism **616***a* may engage with the top surface **441** of the panel portion **442** to urge the module frame **112***a* towards the upper portion **614**. In addition, the portion **445** may be sandwiched between the hooked mechanism **616***a* and the upper portion **614**. In some embodiments, the hooked mechanism **616***a* may prevent the module frame **112***a* from moving too far along the width of the hooked mounting rail **602** (e.g., moving too far right in FIG. **6**B) to facilitate alignment of the PV module **108***a*. The process to couple the PV module **108***a* to the additional attachment mounting rail is discussed in more detail below.

[0093] During installation of the PV module **108***b*, the PV module **108***b* may be positioned such that the bottom surface **122***b* physically engages with the upper portion **614**. In addition, the PV module **108***b* may be positioned such that the portion **445** of the module frame **112***b* is proximate to the aperture **630***b*. The PV module **108***b* may be moved such that the module frame **112***b* moves along the width of the hooked mounting rail **602** (e.g., along a left direction in FIG. **6**B) to cause the portion **445** of the module frame **112***b* to enter the aperture **630***b* and the hooked mechanism **616***b* to engage with the portion **445** of the module frame **112***b*. In particular, the hooked mechanism **616***b* may engage with the top surface **441** of the panel portion **442** to urge the module frame **112***b* towards the upper portion **614**. In some embodiments, the hooked mechanism **616***b* may prevent the module frame **112***b* from moving too far along the width of the hooked mounting rail **602** (e.g., moving too far left in FIG. **6**B) to facilitate alignment of the PV module **108***b*. The process to couple the PV module **108***b* to the attachment mounting rail **604** is discussed in more detail below.

[0094] In this manner, the hooked mounting rail **602** couples to the PV modules **108***a-b* via the module frames **112***a-b* and the hooked mechanisms **616***a-b* without the use of any fasteners or tools (e.g. the toolless process). Sequences of movements to interface an example PV module with example mounting rails are described in more detail below in relation to FIG. **11**.

[0095] During installation of the PV module **108***c*, a similar process may be followed to couple the PV module **108***c* to the additional hooked mounting rail as followed to couple the PV module **108***a* to the hooked mounting rail **602** except with relation to the PV module **108***c* and the additional hooked mounting rail.

[0096] The upper portion **614** includes the surface **624**, that physically engages with the bottom surfaces **122***a-b* of the module frames **112***a-b*. In some embodiments, the surface **624** may include a first part **735** (shown in FIGS. **7**A and **7**C) and a second part **737** (shown in FIGS. **7**A and **7**C). The first part **735** may physically engage with the bottom surface **122***a* of the module frame **112***a*. The second part **737** may physically engage with the bottom surface **122***b* of the module frame **112***b*.

[0097] Each of the hooked mechanisms **616***a-b* may include a first component **731***a-b* (generally referred to in the present disclosure as the first component **731**), a second component **733***a-b* (generally referred to in the present disclosure as the second component **733**), or both. The first component **731** may extend from the surface **624** of the upper portion **614**. The second component **733** may extend from the first component **731** and towards the upper portion **614** such that the second component **733** includes a radius of curvature. The first component **731**, the second component **733**, or both may at least partially define the apertures **630***a-b*. The second component **733** may extend in a direction towards the upper portion **614** to apply a force on the top surface **441** of the corresponding panel portion **442** when engaged with the module frames **112***a-b*.

[0098] The hooked mechanisms **616***a-b*, when engaged with the panel portions **442**, may apply force on the top surface **441** to urge the corresponding module frames **112***a-b* towards the upper portion **614**. For example, the portion **445** of the module frame **112***a* entering the aperture **630***a* may cause the hooked mechanism **616***a* to selectively deform or move to cause the hooked mechanism **616***a* to apply the force on the top surface **441** of the panel portion **442**. As another example, the portion **445** of the module frame **112***b* entering the aperture **630***b* may cause the hooked mechanism **616***b* to selectively deform or move to cause the hooked mechanism **616***b* to apply the force on the top surface **441** of the panel portion **442**. In this way, the hooked mechanisms **616***a-b* may prevent the bottom surfaces **122***a-b* from unintentionally disengaging from the upper portion **614**.

[0099] The hooked mechanisms **616***a-b* are illustrated in FIGS. **6**A-**7**E as extending along an entire length of the hooked mounting rail **602** for example purposes. It is understood that the hooked mechanisms **616***a-b* may be separated into any appropriate number of separate portions. For example, the hooked mechanisms **616***a-b* may be separated into two, three, four, or more separate portions.

[0100] In some embodiments, the hooked mechanisms **616***a-b* may align the PV modules **108***a-b* relative to the upper portion **614**. In addition, the hooked mechanisms **616***a-b* may be sized to prevent the PV modules **108***a-b* from contacting each other. Further, the hooked mechanism **616***a-b* may physically engage with the portions **445** of the corresponding module frames **112***a-b* to prevent the module frames **112***a-b* from moving too far along the width of the hooked mounting rail **602**.

[0101] In some embodiments, the mounting rail **402** may include the retention member **118**. In other embodiments, the retention member **118** may be omitted.

[0102] In some embodiments, the hooked mounting rail **602** may include a single unitary piece of material in which the hooked mechanisms **616***a-b* are punched or otherwise formed from. In other embodiments, the hooked mounting rail **602** may include multiple pieces of material. The hooked mechanism **616***a-b* may be connected to the hooked mounting rail **602** via a weld, a rivet, a threaded fastener, an adhesive, a clinch joint, or a snap-in feature. In these and other embodiments, the hooked mounting rail **602** may include aluminum, steel, or any other appropriate material. In some embodiments, the hooked mounting rail **602** may be formed using an extrusion process.

[0103] With combined reference to FIGS. **6**A and **6**B and FIG. **8**, the attachment feature **605** may couple the PV modules **108***b-c* to the attachment mounting rail **604** using fasteners **671** (shown in FIGS. **6**A and **6**B) (e.g., a tooled process). In some embodiments, the attachment feature **605** may couple the PV modules **108***b-c* to the attachment mounting rail **604** in a way that is different than how the hooked mechanisms **616***a-b* couple the PV module **108***a-b* to the hooked mounting rail **602**.

[0104] In some embodiments, the attachment feature **605** may include one or more openings **853** and/or **877** defined by an upper portion **814** (shown in FIG. **8**) of the attachment mounting rail **604**. Additionally or alternatively, the fasteners **671** may include the bolts **477**, the nuts **473**, clamps **675**, or some combination thereof. The openings **853** may align with the openings **129** defined by the return flanges **120***c-d* of the module frames **112***c-d*. Additionally or alternatively, the openings **877** may be positioned between the PV modules **108***c-d* when they are engaged with the attachment mounting rail **604**.

[0105] The openings **853** of the attachment feature **605** and the openings **129** defined by the return flanges **120***c-d* (examples of which are shown in FIG. **6**A in relation to the return flanges **120***a-b* of the module frames **112***a-b*) may receive the fasteners **671** to couple the PV modules **108***c-d* to the attachment mounting rail **604**. Additionally or alternatively, the openings **877** may receive the fasteners **671** to couple the PV modules **108***c-d* to the attachment mounting rail **604**.

[0106] During installation of the PV modules **108**c-d, the PV modules **108**c-d may be positioned such that the bottom surfaces **122**c-d of the module frames **112**c-d physically engage with the upper portion **814**. In addition, during installation, the PV modules **108**c-d may be positioned such that the openings **853** of the attachment feature **605** are aligned with the openings **129** defined by the return flanges **120**c-d. Further, during installation, the PV modules **108**c-d may be positioned such that the openings **877** are between the PV module **108**c-d. The openings **853** and/or **877** of the attachment feature **605** and the openings **129** defined by the return flanges **120**c-d may receive the fasteners **671**. The fasteners **671** may draw the module frames **112**c-d towards the upper portion **814** and couple the module frames **112**c-d to the attachment mounting rail **604**. For example, as shown, the fasteners **671** include the bolts **477** and the nuts **473**, which interface with each other to draw the module frames **112**a-b towards the upper portion **814**. As another example, as shown, the clamps **675** include top-down clamps configured to draw the module frames **112**a-b towards the upper portion **814** and couple the PV modules **108**c-d to the attachment mounting rail **604**. The clamps **675** as shown include bolts **681** that are received by the openings **877** and that interface with washer portions **683** and nuts **679** (single instance of the nuts is shown in FIG. **6**B). The bolts **681** may interface with the washer portions **683** and the nuts **679** to draw the washer portions **683** towards the PV modules **108**c-d and subsequently the module frames **112**c-d towards the upper portion **814**. In some embodiments, the openings **853** and/or **877** may include threaded portions configured to interface with threaded portions of the bolts **477** or the bolts **681**.

[0107] In this manner, the attachment mounting rail **604** couples to the PV modules **108**c-d via the module frames **112**c-d and the fasteners **671**. The attachment feature **605** is illustrated and described as including the openings **853**, the openings **877**, and the fasteners **671** for example purposes. It is understood that one or more parts of the attachment feature **605** may be omitted. For example, the openings **877** and the clamps **675** may be omitted. As another example, the openings **853**, the bolts **477**, and the nuts **473** may be omitted. It is also understood that the attachment feature **605** may include any appropriate device to couple the PV modules **108**c-d to the attachment mounting rail **604**. For example, the attachment feature **605** may include bolts, nuts, threaded fasteners, thru-bolts, clips, top-down clamps, rivets, or some combination thereof.

[0108] During installation of the PV module **108**a, a similar process may be followed to couple the PV module **108**a to the additional attachment mounting rail as followed to couple the PV module **108**b to the attachment mounting rail **604** except with relation to the PV module **108**a and the additional attachment mounting rail.

[0109] In some embodiments, the hooked mounting rail **602** may include multiple instances of the hooked mechanism **116** discussed above in relation to FIGS. **1**A-**2**G instead of the hooked mechanisms **616**a-b.

[0110] FIG. **9** illustrates an example module frame **112**. The module frame **112** may correspond to the module frames **112**a-b of FIGS. **1**A-**1**C, **3**, **4**A-**4**C, and **6**A-**6**C. The module frame **112** may include a sidewall **936** that is connected to the return flange **120**. The return flange **120** may extend from the sidewall **936** and include the top surface **123** and the bottom surface **122**. In addition, the module frame **112** may include the sidewall **438** that is connected to the panel portion **442**, an upper portion **944**, and a base portion **446**. The upper portion **944** and the base portion **446** may be connected to the sidewall **936** to connect the sidewall **438** to the sidewall **936**. In addition, the sidewall **438**, the panel portion **442**, and the upper portion **944** may define a panel opening **940** configured to receive the PV panels **110**.

[0111] Referring to FIG. **10**, an example torque tube interface **1000** including example mounting rails **1002**a-b to couple an example PV module **108** is shown. The torque tube interface **1000** may include the mounting rails **1002**a-b (generally referred to in the present disclosure as mounting rail **1002**) that couple to different sides of the PV module **108**. The torque tube interface **1000** may be implemented to couple the PV module **108** to any appropriate solar component such as the torque tube **106** (not shown in FIG. **10**).

[0112] In some embodiments, the mounting rail **1002** may include an asymmetrical configuration to couple to different PV modules **108** in different ways. For example, as discussed in more detail below, the mounting rail **1002**a may couple to the PV module **108** via an attachment feature **1005** and the mounting rail **1002**b may couple to the PV module **108** via a hooked mechanism **1016**.

[0113] The mounting rail **1002** may include an upper portion **1014**. The upper portion **1014** may engage with the PV module **108**. For example, the upper portion **1014** may interface with the bottom surface **122** of the module frame **112** as described elsewhere. In addition, the hooked mechanism **1016** may at least partially define one or more apertures **1030** configured to receive a portion of the PV module **108**. For example, the portion **445** of the module frame **112**a described elsewhere. The aperture **1030** may receive the PV module **108** to position at least part PV module **108** between the hooked mechanism **1016** and the upper portion **1014**.

[0114] The aperture **1030** may receive the PV module **108** to couple the PV module **108** to the mounting rail **1002**. In addition, the hooked mechanism **1016** may physically engage with the PV module **108** to couple the PV module **108** to the mounting rail **1002**. For example, the hooked mechanism **1016** may physically engage with the top surface **441** of the panel portion **442** as described elsewhere. The aperture **1030** may receive the PV module **108** to prevent the PV module **108** from unintentionally disengaging from the upper portion **1014**. Further, the hooked mechanism **1016** may engage with the PV module **108** to maintain the engagement between the upper portion **1014** and the PV module **108**. In other words, the hooked mechanism **1016** and the upper portion **1014** may sandwich part of the PV module **108** between the hooked mechanism **1016** and the upper portion **1014** to prevent the PV module **108** from moving away from a surface **1024** of the upper portion **1014** (e.g., moving in an upward direction in FIG. **10**).

[0115] The hooked mechanism **1016** may selectively deform to permit the part of the PV module **108** to enter the aperture **1030**. The hooked mechanism **1016** may include a first component **1031** connected to the upper portion **1014** and a second component **1026**. The second component **1026** may interface with the PV module **108** during installation to cause the hooked mechanism **1016** to selectively deform.

[0116] During installation, the PV module **108** may be positioned proximate to the second component **1026**. The PV module **108** may be moved toward the second component **1026** (e.g., a downward direction in FIG. **10**) to cause the PV module **108** to interface with the second component **1026**. The PV module **108** may continue to be moved to cause the hooked mechanism **1016** to selectively deform. The hooked mechanism **1016** may selectively deform and move to transition from a first position (e.g., an equilibrium position as shown in FIG. **10**) to a second position. In the second position, the PV module **108** may move along a surface of the second component **1026** and be received by and be at least partially disposed in the aperture **1030**. In other words, the PV module **108** engaging with the second component **1026** may cause the second component **1026** to move out of the way of the PV module **108** and permit the PV module **108** to be received by the aperture **1030**.

[0117] When the PV module **108** is disposed within the aperture **1030**, the hooked mechanism **1016** may return to the first position or transition to a third position such that a portion of the PV module **108** is positioned between the hooked mechanism **1016** and the upper portion **1014** to interlock the PV module **108** and the mounting rail **1002**. For example, the second component **1026** and the upper portion **1014** may sandwich part of the PV module **108**.

[0118] As shown in FIG. **10**, the attachment feature **1005** includes the bolts **477**. The attachment feature **1005** may also include the nuts **473** (not shown in FIG. **10**) or any other appropriate device to interface with the bolts **477**. The attachment feature **1005** couples the PV module **108** to the mounting rail **1002** in the tooled process. In other words, the attachment feature **1005** may couple the PV module **108** to the mounting rail **1002** in a way that is different than how the hooked mechanism **1016** couples the PV module **108** to the mounting rail **1002**.

[0119] In some embodiments, the attachment feature **1005** may include one or more openings (not shown in FIG. **10**) defined by the upper portion **1014**. The openings defined by the upper portion **1014** may align with openings (not shown in FIG. **10**) defined by the PV module **108** (e.g., the openings **129** defined by the return flange **120** of the module frame **112**). The openings of the attachment feature **1005** and the openings defined by the PV module **108** may receive the bolts **477** to couple the PV module **108** to the mounting rail **1002**.

[0120] As discussed above, during installation, the PV module **108** may be moved such that a portion of the PV module **108** is received by the aperture **1030**. When the portion of the PV module **108** is positioned within the aperture **1030**, the PV module **108** may physically engage with the upper portion **1014** (e.g., the bottom surface **122*b*** of the module frame **112*b*** physically engages with the upper portion **1014**). In addition, the PV module **108** may be positioned such that the openings of the attachment feature **1005** are aligned with the openings **129** defined by the PV module **108**. The openings of the attachment feature **1005** and the openings **129** defined by the PV module **108** may receive the bolts **477**. The bolts **477** may interface with the nuts **473** to draw the PV module **108** towards the upper portion **1014** and couple the PV module **108** to the mounting rail **1002**.

[0121] In this manner, the mounting rail **1002** couples to the PV module **108** via the attachment feature **1005**. The attachment feature **1005** is illustrated and described as including the bolts **477**, the nuts **473**, and the openings for example purposes. The attachment feature **1005** may include any appropriate device to couple the PV module **108** to the mounting rail **1002**. For example, the attachment feature **1005** may include bolts, nuts, threaded fasteners, thru-bolts, clips, top-down clamps, rivets, or some combination thereof.

[0122] The hooked mechanism **1016** may include a single portion or multiple portions that are connected to the upper portion **1014** and positioned at different locations on the upper portion **1014**. For example, the hooked mechanism **1016** may include one, two, three, or more separate portions. In some embodiments, the hooked mechanism **1016** may align the PV module **108** relative to the upper portion **1014**.

[0123] In some embodiments, the mounting rail **1002** may include one or more stopper devices **1011**. The stopper devices **1011** may be connected to and extend from the upper portion **1014**. In some embodiments, the stopper devices **1011** may be positioned proximate to and extend generally parallel to the hooked mechanism **1016**. The stopper devices **1011** may interface with the hooked mechanism **1016** during installation of the PV module **108** to prevent the hooked mechanism **1016** from selectively deforming beyond a pre-determined threshold. In some embodiments, the pre-determined threshold may be equal to or greater than eight degrees from an equilibrium position of the hooked mechanism **1016**. The hooked mechanism **1016** is illustrated in FIG. **10** in the equilibrium position.

[0124] In some embodiments, the mounting rail **1002** may include the retention member **118**. In other embodiments, the retention member **118** may be omitted.

[0125] In some embodiments, the mounting rail **1002** may include a single unitary piece of material in which the hooked mechanism **1016**, the stopper devices **1011**, or both are punched or otherwise formed from. In other embodiments, the mounting rail **1002** may include multiple pieces of material. The hooked mechanism **1016**, the stopper devices **1011**, or both may be connected to the mounting rail **1002** via a weld, a rivet, a threaded fastener, an adhesive, a clinch joint, or a snap-in feature In these and other embodiments, the mounting rail **1002** may include aluminum, steel, or any other appropriate material. In some embodiments, the mounting rail **1002** may be formed using an extrusion process.

[0126] FIG. **11** illustrates graphical representations **1102*a-c*** of sequences of movements to interface an example PV module **108** with different mounting rails described in the present disclosure. The example PV module **108** may correspond to the PV modules **108*a-c*** described elsewhere. The graphical representation **1102*a*** illustrates a sequence of movements to interface the PV module **108** with the mounting rail **102** of FIGS. **1A-2G** or the mounting rail **302** of FIG. **3**. The graphical representation **1102*b*** illustrates a sequence of movements to interface the PV module **108** with the mounting rail **402** of FIGS. **4A-5G** or the torque tube interface **600** of FIGS. **6A-8**. The graphical representation **1102*c*** illustrates a sequence of movements to interface the PV module **108** with the torque tube interface **1000** of FIG. **10**.

[0127] With reference to the graphical representation **1102*a***, a first movement represented by arrow **1101** includes moving the PV module **108** to the side. As shown in FIG. **11**, the first movement **1101** may include moving the PV module **108** to the left. The first movement **1101** may allow a portion **1130** of the PV module **108** (e.g., the return flange **120**) that interfaces with a hooked mechanism (e.g., hooked mechanism **116** and/or hooked mechanism **316**) to clear the hooked mechanism. In some embodiments, the first movement may be omitted. A second movement represented by arrow **1103**

includes moving the PV module **108** down. The second movement **1103** may occur until the PV module **108** engages with an upper portion (e.g., upper portion **114**) of a mounting rail (e.g., mounting rail **102** and/or **302**).

[0128] A third movement represented by arrow **1105** includes moving the PV module **108** to the side in a direction that is different than the first movement **1101**. As shown in FIG. **11**, the third movement **1105** may include moving the PV module **108** to the right. The third movement **1105** may cause the portion **1130** of the PV module **108** to interface with the hooked mechanism. A fourth movement represented by arrow **1107** may include coupling the PV module **108** to the mounting rail via an attachment feature (e.g., the attachment feature **105**). The directions of the first movement **1101** and the third movement **1105** may be reversed (e.g., the first movement **1101** may be to the right and the third movement **1105** may be to the left) to permit the PV module **108** to interface with an inverted arrangement of the hooked mechanism and the attachment feature.

[0129] With reference to graphical representation **1102***b*, a first movement represented by arrow **1109** includes moving the PV module **108** down. The first movement **1109** may occur until the PV module **108** engages with an upper portion (e.g., upper portion **414**, **614**, and/or **814**) of the mounting rail (e.g., mounting rail **402**, **602**, and/or **604**).

[0130] A second movement represented by arrow **1111** includes moving the PV module **108** to a side. As shown in FIG. **11**, the second movement **1111** may include moving the PV module **108** to the left. The second movement **1111** may cause the portion **1130** (e.g., the portion **445** of the module frame **112**) of the PV module **108** to interface with the hooked mechanism (e.g., hooked mechanism **416** and/or **616**). A third movement represented by arrow **1113** may include coupling the PV module **108** to the mounting rail via an attachment feature (e.g., the attachment feature **405** and/or **605**). The direction of the second movement **1111** may be reversed (e.g., the second movement **1111** may be to the right) to permit the PV module **108** to interface with an inverted arrangement of the hooked mechanism and the attachment feature.

[0131] With reference to graphical representation **1102***c*, a first movement represented by arrow **1115** includes moving the PV module **108** down. The first movement **1115** may occur until the PV module **108** engages with the hooked mechanism (e.g., hooked mechanism **1016**) and causes the hooked mechanism to selectively deform and permit the portion **1130** to enter an aperture (e.g., aperture **1030**) defined by the hooked mechanism and an upper portion (e.g., upper portion **1014**) of a mounting rail (e.g., mounting rail **1002**). A second movement represented by arrow **1117** may include coupling the PV module **108** to the mounting rail via an attachment feature (e.g., the attachment feature **1005**).

[0132] Embodiment 1 includes a mounting rail configured to couple to a first photovoltaic (PV) module and a second PV module, the mounting rail comprising: a hooked mechanism configured to: at least partially define an aperture configured to receive a portion of a first module frame associated with the first PV module; and physically engage with a surface of the first module frame to couple the first module frame to the mounting rail and to prevent the first module frame from unintentionally uncoupling from the mounting rail; and an attachment feature configured to interface with a second module frame associated with the second PV module to couple the second module frame to the mounting rail.

[0133] Embodiment 2 includes the mounting rail of embodiment 1, wherein the surface of the first module frame comprises at least one of: a top surface of a return flange of the first module frame and the portion of the first module frame comprises an edge of the return flange; or a top surface of a panel portion of the first module frame and the portion of the first module frame comprises at least one of a sidewall of the first module frame, at least part of the panel portion, or at least part of a base portion of the first module frame.

[0134] Embodiment 3 includes the mounting rail of embodiment 1, wherein the attachment feature: is configured to prevent the second module frame from unintentionally uncoupling from the mounting rail; and comprises at least one of: a fastener; a bolt; a nut; a clip; a threaded fastener; a thru-bolt; a top-down clamp; or a rivet.

[0135] Embodiment 4 includes the mounting rail of embodiment 1, wherein the attachment feature comprises another hooked mechanism.

[0136] Embodiment 5 includes the mounting rail of embodiment 1, wherein the hooked mechanism is configured to couple the first PV module to the mounting rail in a toolless process and the attachment feature is configured to couple the second PV module to the mounting rail using a tooled process.

[0137] Embodiment 6 includes the mounting rail of embodiment 1, wherein the attachment feature is configured to couple the second PV module to the mounting rail via a way that is different than how the hooked mechanism couples the first PV module to the mounting rail.

[0138] Embodiment 7 includes the mounting rail of embodiment 1, wherein the hooked mechanism is connected to at least one of: an edge of the mounting rail; or an upper portion of the mounting rail.

[0139] Embodiment 8 includes the mounting rail of embodiment 1, further comprising a stopper device configured to interface with the hooked mechanism to prevent the hooked mechanism from selectively deforming beyond a pre-determined threshold.

[0140] Embodiment 9 includes the mounting rail of embodiment 1, further comprising a retention member configured to interface with the first module frame, during installation, to cause the retention member to selectively deform to permit the portion of the first module frame to be received by and be at least partially disposed within the aperture.

[0141] Embodiment 10 includes the mounting rail of embodiment 9, wherein the retention member is configured to be received by an opening defined by the first module frame to facilitate a position of the first module frame relative to the mounting rail and to prevent the portion of the first module frame from unintentionally exiting the aperture.

[0142] Embodiment 11 includes the mounting rail of embodiment 1, wherein: the hooked mechanism comprises a first hooked mechanism; the mounting rail further comprises a second hooked mechanism; the first hooked mechanism is positioned proximate a first longitudinal end of the mounting rail; and the second hooked mechanism is positioned proximate a second longitudinal end of the mounting rail.

[0143] Embodiment 12 includes the mounting rail of embodiment 1 further comprising an upper portion comprising a surface configured to physically engage with a first bottom surface of the first module frame and a second bottom surface of the second module frame, wherein the hooked mechanism and the upper portion together at least partially define the aperture.

[0144] Embodiment 13 includes the mounting rail of embodiment 12, wherein: the hooked mechanism overlaps a portion of the surface of the upper portion to at least partially define the aperture; and the hooked mechanism and the upper portion together are configured to sandwich part of the first module frame between the hooked mechanism and the upper portion.

[0145] Embodiment 14 includes the mounting rail of embodiment 12, wherein: the surface of the upper portion comprises a first part and a second part; the first part is configured to physically engage with the first bottom surface; and the second part is configured to physically engage with the second bottom surface to permit the second PV module to be positioned proximate to the first PV module.

[0146] Embodiment 15 includes the mounting rail of embodiment 1, wherein the hooked mechanism is positioned at a longitudinal end of the mounting rail.

[0147] Embodiment 16 includes the mounting rail of embodiment 1, wherein the hooked mechanism is positioned a distance from a longitudinal end of the mounting rail.

[0148] Embodiment 17 includes the mounting rail of embodiment 1, wherein the hooked mechanism extends continuously along at least part of a length of the mounting rail.

[0149] Embodiment 18 includes the mounting rail of embodiment 1, wherein the hooked mechanism comprises a plurality of hooked mechanisms positioned at different locations of the mounting rail.

[0150] Embodiment 19 includes the mounting rail of embodiment 18, further comprising a plurality of retention members, wherein each retention member of the plurality of retention members is positioned between two hooked mechanisms of the plurality of hooked mechanisms.

[0151] Embodiment 20 includes the mounting rail of embodiment 1, wherein the mounting rail comprises a single unitary piece of material.

[0152] Embodiment 21 includes a mounting rail configured to couple to a first photovoltaic (PV) module and a second PV module, the mounting rail comprising: a first hooked mechanism configured to: at least partially define a first aperture configured to receive a portion of a first module frame associated with the first PV module; and physically engage with a surface of the first module frame to couple the first module frame to the mounting rail and to prevent the first module frame from unintentionally uncoupling from the mounting rail; and a second hooked mechanism configured to: at least partially define a second aperture configured to receive a portion of a second module frame associated with the second module frame to couple the second module frame to the mounting rail; and physically engage with a surface of the second module frame to prevent the second module frame from unintentionally uncoupling from the mounting rail.

[0153] Embodiment 22 includes the mounting rail of embodiment 21, wherein: the surface of the first module frame comprises at least one of: a top surface of a return flange of the first module frame and the portion of the first module frame comprises an edge of the return flange; or a top surface of a panel portion of the first module frame and the portion of the first module frame comprises at least one of a sidewall of the first module frame, at least part of the panel portion of the first module frame, or at least part of a base portion of the first module frame; and the surface of the second module frame comprises at least one of: a top surface of a return flange of the second module frame and the portion of the second module frame comprises an edge of the return flange; or a top surface of a panel portion of the second module frame and the portion of the second module frame comprises at least one of a sidewall of the second module frame, at least part of the panel portion of the second module frame, or at least part of a base portion of the second module frame.

[0154] Embodiment 23 includes the mounting rail of embodiment 21, wherein at least one of: the first hooked mechanism is connected to an edge of the mounting rail and the second hooked mechanism is connected to an upper portion of the mounting rail; the first hooked mechanism is connected to the upper portion of the mounting rail and the second hooked mechanism is connected to the edge of the mounting rail; the first hooked mechanism is connected to a first edge of the mounting rail and the second hooked mechanism is connected to a second edge of the mounting rail; or the first hooked mechanism is connected to the upper portion of the mounting rail and the second hooked mechanism is connected to the upper portion of the mounting rail.

[0155] Embodiment 24 includes the mounting rail of embodiment 21, further comprising a stopper device configured to interface with the first hooked mechanism to prevent the first hooked mechanism from selectively deforming beyond a pre-determined threshold.

[0156] Embodiment 25 includes the mounting rail of embodiment 21 further comprising a retention member configured to interface with the first module frame, during installation, to cause the retention member to selectively deform to permit the portion of the first module frame to be received by and be at least partially disposed within the first aperture.

[0157] Embodiment 26 includes the mounting rail of embodiment 25, wherein the retention member is configured to be received by an opening defined by the first module frame to facilitate a position of the first module frame relative to the mounting rail and to prevent the portion of the first module frame from unintentionally exiting the first aperture.

[0158] Embodiment 27 includes the mounting rail of embodiment 21, wherein: the first hooked mechanism is positioned proximate a first longitudinal end of the mounting rail; and the second hooked mechanism is positioned proximate a second longitudinal end of the mounting rail.

[0159] Embodiment 28 includes the mounting rail of embodiment 21 further comprising an upper portion comprising a surface configured to physically engage with a first bottom surface of the first module frame and a second bottom surface of the second module frame, wherein the first hooked mechanism and the upper portion together at least partially define the first aperture and the second hooked mechanism and the upper portion at least partially define the second aperture.

[0160] Embodiment 29 includes the mounting rail of embodiment 28, wherein: the first hooked mechanism overlaps a portion of the surface of the upper portion to at least partially define the first aperture; and the first hooked mechanism and the upper portion together are configured to sandwich part of the first module frame between the first hooked mechanism and the upper portion.

[0161] Embodiment 30 includes the mounting rail of embodiment 28, wherein: the surface of the upper portion comprises a first part and a second part; the first part is configured to physically engage with the first bottom surface; and the second part is configured to physically engage with the second bottom surface to permit the second PV module to be positioned proximate to the first PV module.

[0162] Embodiment 31 includes the mounting rail of embodiment 21, wherein the first hooked mechanism is positioned at a longitudinal end of the mounting rail.

[0163] Embodiment 32 includes the mounting rail of embodiment 21, wherein the first hooked mechanism is positioned a distance from a longitudinal end of the mounting rail.

[0164] Embodiment 33 includes the mounting rail of embodiment 21, wherein the first hooked mechanism extends continuously along at least part of a length of the mounting rail.

[0165] Embodiment 34 includes the mounting rail of embodiment 21, wherein the first hooked mechanism comprises a plurality of hooked mechanisms positioned at different locations of the mounting rail.

[0166] Embodiment 35 includes the mounting rail of embodiment 34, further comprising a plurality of retention members, wherein each retention member of the plurality of retention members is positioned between two hooked mechanisms of the plurality of hooked mechanisms.

[0167] Embodiment 36 includes the mounting rail of embodiment 21, wherein the mounting rail comprises a single unitary piece of material.

[0168] Embodiment 37 includes a system configured to couple to a photovoltaic (PV) module, the system comprising: a torque tube; a hooked mounting rail configured to interface with the torque tube at a first location, the hooked mounting rail comprising a hooked mechanism configured to: at least partially define an aperture configured to receive a portion of a first part of a module frame associated with the PV module to couple the module frame to the hooked mounting rail; and physically engage with a surface of the first part of the module frame to prevent the module frame from unintentionally uncoupling from the hooked mounting rail; and an attachment mounting rail configured to interface with the torque tube at a second location, a distance between the first location and the second location corresponding to a width of the PV module, the attachment mounting rail comprising an attachment feature configured to interface with a second part of the module frame to couple the module frame to the attachment mounting rail, the second part of the module frame being positioned on an opposite side of the module frame as the first part.

[0169] Embodiment 38 includes the system of embodiment 37, wherein: the PV module comprises a first PV module, the hooked mechanism comprises a first hooked mechanism, and the attachment feature comprises a first attachment feature; the hooked mounting rail comprises a second hooked mechanism configured to: at least partially define an aperture configured to receive a portion of a first part of a module frame associated with a second PV module to couple the module frame associated with the second PV module to the hooked mounting rail; and physically engage with a surface of the first part of the module frame associated with the second PV module to prevent the module frame associated with the second PV module from unintentionally uncoupling from the hooked mounting rail; and the system further comprises an additional attachment mounting rail configured to interface with the torque tube at a third location, the third location being located on an opposite side of the hooked mounting rail as the second location, the additional attachment mounting rail comprising a second attachment feature configured to interface with a second part of the module frame associated with the second PV module to couple the module frame associated with the second PV module to the additional attachment mounting rail.

[0170] Embodiment 39 includes the system of embodiment 37, wherein the surface of the module frame associated with the PV module comprises at least one of: a top surface of a return flange of the module frame and the portion of the module frame comprises an edge of the return flange; or a top surface of a panel portion of the module frame and the portion of the module frame comprises at least one of a sidewall of the module frame, at least part of the panel portion, or at least part of a base portion of the module frame.

[0171] Embodiment 40 includes the system of embodiment 37, wherein the attachment feature: is configured to prevent the second module frame from unintentionally uncoupling from the mounting rail; and comprises at least one of: a fastener; a bolt; a nut; a clip; a threaded fastener; a thru-bolt; a top-down clamp; or a rivet.

[0172] Embodiment 41 includes the system of embodiment 37, wherein the attachment feature comprises another hooked mechanism.

[0173] Embodiment 42 includes the system of embodiment 38, wherein the hooked mechanism is configured to couple the first PV module to the mounting rail in a toolless process and the attachment feature is configured to couple the second PV module to the mounting rail using a tooled process.

[0174] Embodiment 43 includes the system of embodiment 38, wherein the attachment feature is configured to couple the second PV module to the mounting rail via a way that is different than how the hooked mechanism couples the first PV module to the mounting rail.

[0175] Embodiment 44 includes the system of embodiment 37, further comprising a stopper device configured to interface with the hooked mechanism to prevent the hooked mechanism from selectively deforming beyond a pre-determined threshold.

[0176] Embodiment 45 includes the system of embodiment 37, wherein the hooked mounting rail further comprises a retention member configured to interface with the module frame, during installation, to cause the retention member to selectively deform to permit the portion of the module frame to be received by and be at least partially disposed within the aperture.

[0177] Embodiment 46 includes the system of embodiment 45, wherein the retention member is configured to be received by an opening defined by the module frame to facilitate a position of the module frame relative to the hooked mounting rail and to prevent the portion of the module frame from unintentionally exiting the aperture.

[0178] Embodiment 47 includes the system of embodiment 37, wherein the hooked mounting rail further comprises an upper portion comprising a surface configured to physically engage with a bottom surface of the module frame and the hooked mechanism and the upper portion together at least partially define the aperture.

[0179] Embodiment 48 includes the system of embodiment 47, wherein: the hooked mechanism overlaps a portion of the surface of the upper portion to at least partially define the aperture; and the hooked mechanism and the upper portion together are configured to sandwich part of the module frame between the hooked mechanism and the upper portion.

[0180] Embodiment 49 includes a mounting rail configured to couple to a first photovoltaic (PV) module and a second PV module, the mounting rail comprising: a hooked mechanism configured to: at least partially define an aperture configured to receive a portion of a first module frame associated with the first PV module to couple the first module frame to the mounting rail; interface with the first module frame during installation of the first PV module to cause the hooked mechanism to selectively deform and permit at least a portion of the first module frame to be received by and be at least partially disposed in the aperture to interlock the first module frame with the mounting rail; and physically engage with a top surface of a panel portion of the first module frame to prevent the first module frame from unintentionally uncoupling from the mounting rail; and an attachment feature configured to interface with a second module frame associated with the second PV module to couple the second module frame to the mounting rail.

[0181] Embodiment 50 includes the mounting rail of embodiment 49, wherein the attachment feature: is configured to prevent the second module frame from unintentionally uncoupling from the mounting rail; and comprises at least one of: a fastener; a bolt; a nut; a clip; a threaded fastener; a thru-bolt; a top-down clamp; or a rivet.

[0182] Embodiment 51 includes the mounting rail of embodiment 49, wherein the attachment feature comprises another hooked mechanism.

[0183] Embodiment 52 includes the mounting rail of embodiment 49, wherein the hooked mechanism is configured to couple the first PV module to the mounting rail in a toolless process and the attachment feature is configured to couple the second PV module to the mounting rail using a tooled process.

[0184] Embodiment 53 includes the mounting rail of embodiment 49, wherein the attachment feature is configured to couple the second PV module to the mounting rail via a way that is different than how the hooked mechanism couples the first PV module to the mounting rail.

[0185] Embodiment 54 includes the mounting rail of embodiment 49 comprising a stopper device configured to prevent movement of the second module frame along a width of the mounting rail.

[0186] Embodiment 55 includes the mounting rail of embodiment 49 comprising a stopper device configured to interface with the hooked mechanism to prevent the hooked mechanism from selectively deforming beyond a pre-determined threshold.

What is claimed is:

1. A mounting rail configured to couple to a first photovoltaic (PV) module and a second PV module, the mounting rail comprising:
- a hooked mechanism configured to:
  - at least partially define an aperture configured to receive a portion of a first module frame associated with the first PV module; and
  - physically engage with a surface of the first module frame to couple the first module frame to the mounting rail and to prevent the first module frame from unintentionally uncoupling from the mounting rail; and
- an attachment feature configured to interface with a second module frame associated with the second PV module to couple the second module frame to the mounting rail.

2. The mounting rail of claim **1**, wherein the surface of the first module frame comprises at least one of:
- a top surface of a return flange of the first module frame and the portion of the first module frame comprises an edge of the return flange; or
- a top surface of a panel portion of the first module frame and the portion of the first module frame comprises at least one of a sidewall of the first module frame, at least part of the panel portion, or at least part of a base portion of the first module frame.

3. The mounting rail of claim **1**, wherein the attachment feature:
- is configured to prevent the second module frame from unintentionally uncoupling from the mounting rail; and
- comprises at least one of:
  - a fastener;
  - a bolt;
  - a nut;
  - a clip;
  - a threaded fastener;
  - a thru-bolt;
  - a top-down clamp; or
  - a rivet.

4. The mounting rail of claim **1**, wherein the attachment feature comprises another hooked mechanism.

5. The mounting rail of claim **1**, wherein the hooked mechanism is configured to couple the first PV module to the mounting rail in a toolless process and the attachment feature is configured to couple the second PV module to the mounting rail using a tooled process.

6. The mounting rail of claim **1**, wherein the attachment feature is configured to couple the second PV module to the mounting rail via a way that is different than how the hooked mechanism couples the first PV module to the mounting rail.

7. The mounting rail of claim **1**, wherein the hooked mechanism is connected to at least one of:
- an edge of the mounting rail; or
- an upper portion of the mounting rail.

8. The mounting rail of claim **1**, further comprising a stopper device configured to interface with the hooked mechanism to prevent the hooked mechanism from selectively deforming beyond a pre-determined threshold.

9. The mounting rail of claim **1**, further comprising a retention member configured to interface with the first module frame, during installation, to cause the retention member to selectively deform to permit the portion of the first module frame to be received by and be at least partially disposed within the aperture.

10. The mounting rail of claim **9**, wherein the retention member is configured to be received by an opening defined by the first module frame to facilitate a position of the first module frame relative to the mounting rail and to prevent the portion of the first module frame from unintentionally exiting the aperture.

11. The mounting rail of claim **1**, wherein:

the hooked mechanism comprises a first hooked mechanism;

the mounting rail further comprises a second hooked mechanism;

the first hooked mechanism is positioned proximate a first longitudinal end of the mounting rail; and

the second hooked mechanism is positioned proximate a second longitudinal end of the mounting rail.

12. The mounting rail of claim **1** further comprising an upper portion comprising a surface configured to physically engage with a first bottom surface of the first module frame and a second bottom surface of the second module frame, wherein the hooked mechanism and the upper portion together at least partially define the aperture.

13. The mounting rail of claim **12**, wherein:

the hooked mechanism overlaps a portion of the surface of the upper portion to at least partially define the aperture; and

the hooked mechanism and the upper portion together are configured to sandwich part of the first module frame between the hooked mechanism and the upper portion.

14. The mounting rail of claim **12**, wherein:

the surface of the upper portion comprises a first part and a second part;

the first part is configured to physically engage with the first bottom surface; and

the second part is configured to physically engage with the second bottom surface to permit the second PV module to be positioned proximate to the first PV module.

15. The mounting rail of claim **1**, wherein the hooked mechanism is positioned at a longitudinal end of the mounting rail.

16. The mounting rail of claim **1**, wherein the hooked mechanism is positioned a distance from a longitudinal end of the mounting rail.

17. The mounting rail of claim **1**, wherein the hooked mechanism extends continuously along at least part of a length of the mounting rail.

18. The mounting rail of claim **1**, wherein the hooked mechanism comprises a plurality of hooked mechanisms positioned at different locations of the mounting rail.

19. The mounting rail of claim **18**, further comprising a plurality of retention members, wherein each retention member of the plurality of retention members is positioned between two hooked mechanisms of the plurality of hooked mechanisms.

20. A system comprising:

a first photovoltaic (PV) module;

a second PV module;

a torque tube; and

a torque tube interface comprising:

a mounting clamp coupled to the torque tube; and

a mounting rail coupled to the first PV module, the second PV module, and the mounting clamp, the mounting rail comprising:

a hooked mechanism:

at least partially defining an aperture receiving a portion of a first module frame associated with the first PV module; and

physically engaged with a surface of the first module frame to couple the first module frame to the mounting rail and to prevent the first module frame from unintentionally uncoupling from the mounting rail; and

an attachment feature interfaced with a second module frame associated with the second PV module to couple the second module frame to the mounting rail.

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